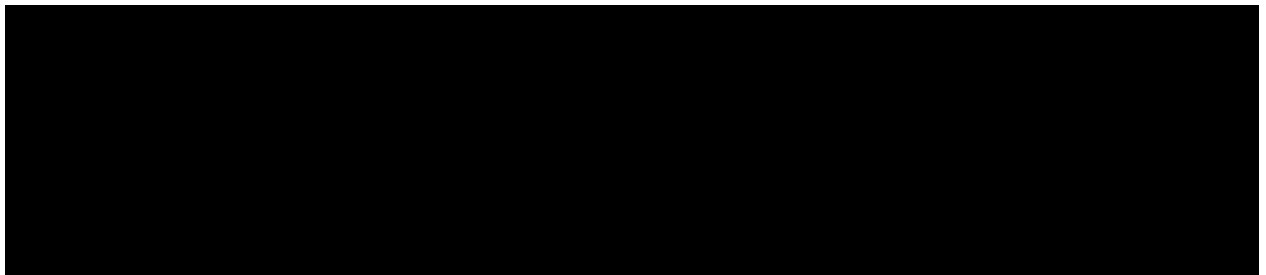
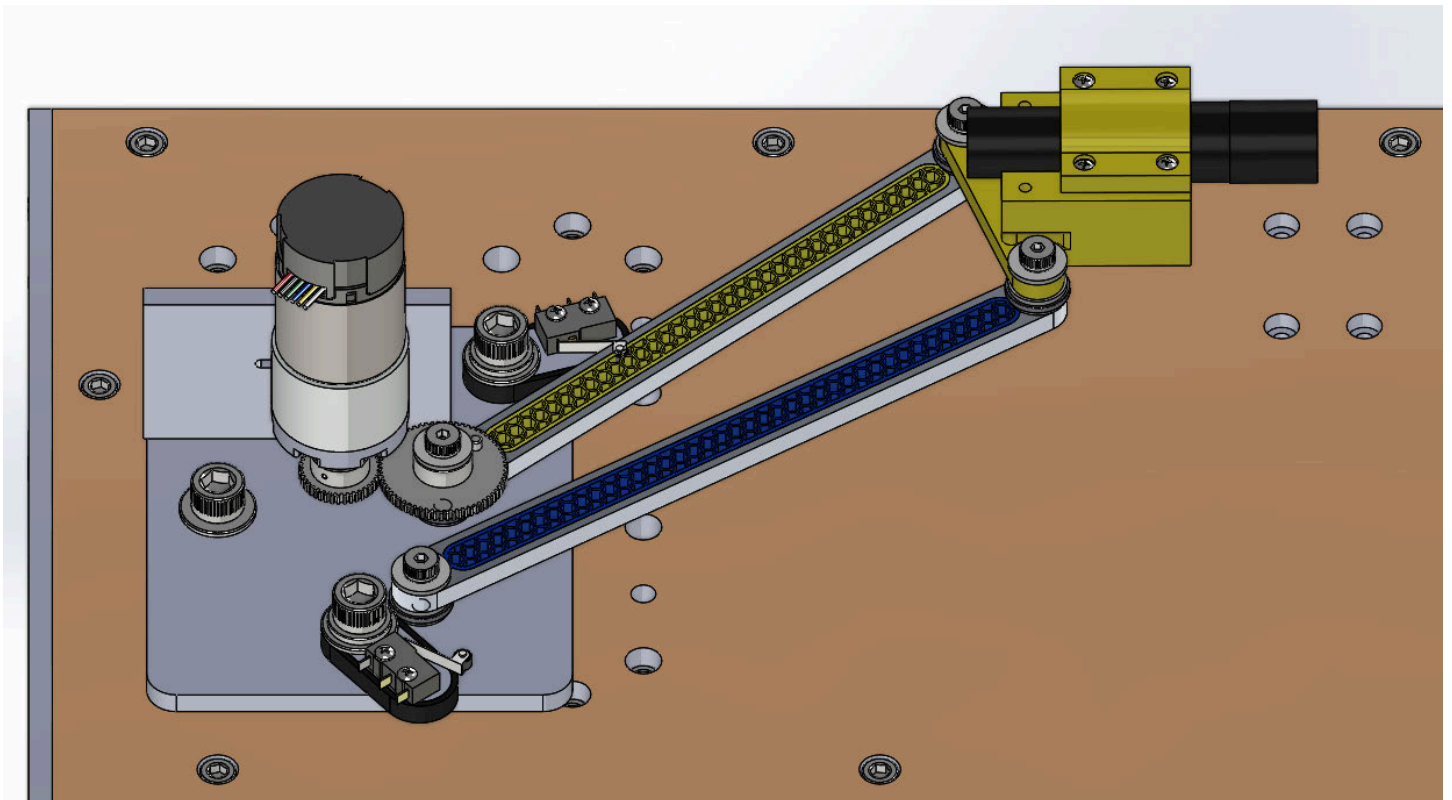


**ME350**  
**Winter 2026 Semester**  
**DESIGN REPORT #2**

Team 41

Rehan Edirisinghe

GSI/IA:



[Introduction \(Reference Only\)](#)

[Audience \(Reference Only\)](#)

[Table & Figure Formatting \(Reference Only\)](#)

[Report Sections \(Reference Only\)](#)

[Due with DR 1 \(Sections 1-3, Appendix A\):](#)

[Section 1: Design Goals, Requirements, and Specifications \(0.5-1 pg\)](#)

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## Section 1: Design Goals, Requirements, and Specifications

### Design Goals

The primary goal of this project is to design and build a mechanism powered by a motor that can automatically detect the target furthest from the proximity sensors. An attached flashlight will be guided towards the photosensors to reset the targets. Once a minimum light requirement will be needed to be incident on the photosensor for, the target will be reset.

### Design Requirements and Specifications

Some of the design requirements and their associated specifications that must be met in order to be successful in this project are:

**Table 1.1** Design Requirements and specifications that we deemed imperative to producing a successful mechanism design.

| Requirement                                                                                                                                        | Specification                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The linkage should not bind or find itself in a toggle position within the operating window.                                                       | The transmission angle deviation must be less than 60° throughout the operating window. This ensures that the motion is predictable, the mechanism will not bind, and joints will not fail.                                                                                                                                     |
| The mechanism's weight should be minimized without compromising its structural integrity.                                                          | The mechanism should employ design solutions that minimize weight where possible, without compromising its operation and durability. This may include some 3D-printed parts, slots, and cutouts. A maximum weight should be 0.75 lbs and be able to withstand a constant point load of 7lbs at 1.5 inches from the ground link. |
| The flashlight should be able to move within the full range of motion of the playing field quickly                                                 | The Flashlight should be able to move from position 1 to 4 within 5 seconds                                                                                                                                                                                                                                                     |
| The mechanism and housing must be manufacturable with the tools available to ME 350 Students.                                                      | The design must be manufacturable using machinery and tools available in the Undergraduate Machine Shop. The surface finish of the final mechanism must be adequate. This includes being sanded, filed, and/or machined to meet the needs of the design.                                                                        |
| The design must be safe to handle by anyone                                                                                                        | No sharp edges and burrs must be adequately deburred. Any exposed bolts should not protrude 1.5x their diameter                                                                                                                                                                                                                 |
| The motor must retain adequate control of the linkage within all points of its specified range of motion                                           | The linkage must respond within 1 second of the proximity sensor being triggered and must move to the specified location within $\pm 0.125'$ .                                                                                                                                                                                  |
| The final design must be aesthetically pleasing and employ design best-practices. Additionally, attention to GD&T and manufacturing best practices | There must be sufficient adjustability and accessibility to critical parts. Excessive use of glue and duct tape must be avoided, and spacers must be a single machined piece rather than multiple washers. The finished product should have parts that are well                                                                 |

|                                                                            |                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| must be employed.                                                          | aligned with each other and minimal play between them.                                                                                                                                                                                              |
| The associated code must be readable, well-structured, and understandable. | Variable names must be purposeful and understandable. Appropriate indentation should be used for functions and nested loops. Comments must be employed to enhance a reader's understanding. The code should be optimized to reduce computing power. |

## Section 2: Design Selection

When selecting the mechanism, we paid attention to the grading criteria and measurable performance metrics outlined in the ME 350 Project description. We also considered the manufacturability, performance, and weight in particular. Given that all 5 of the mechanisms synthesized in Labs 1-3 had a maximum angular deviation of less than 60 degrees from 90 degrees, we decided not to include them in our Pugh Chart. Because of this, we decided to focus on other criteria.

To ensure comparability across all concepts, volume served as a proxy for weight. Our consideration of weight aligns with the rubric, where it accounts for 30% of the grade, and acknowledges that weight influences inertia, which in turn impacts the performance of the system and motors. High weightings were given to the distance between the target and coupler as well as the time taken to complete a movement (compared using a standardized wattage for the motor input). These metrics are likely the most influential factors in the mechanism's success, since both the speed and the amount of light reaching the target directly impact the number of targets successfully sent back over each round. We also considered manufacturability, especially regarding the spacing between ground links, which would affect ease of assembly. This factor received a lower weighting since all proposed link designs were perceived as easily manufacturable, with the only difference being the length of the ground link.

**Table 2.1** Pugh Chart qualitatively represents the criteria we deemed important when choosing from the concept designs. An associated weight was assigned to each criterion based on how important we deemed it to the project's success. The total was calculated by multiplying the ranking by the associated weighting.

| Criterion                                                        | Weight | Option 1 | Option 2 | Option 3 | Option 4 | Option 4 |
|------------------------------------------------------------------|--------|----------|----------|----------|----------|----------|
| Volume                                                           | 3      | 0        | 1        | -1       | 1        | -1       |
| Distance to Targets                                              | 5      | 0        | -1       | -1       | -1       | -1       |
| Time taken to complete the movement with the motor running at 1W | 4      | 0        | -1       | -1       | 1        | -1       |
| Distance between ground links                                    | 2      | 0        | -1       | -1       | -1       | 1        |
|                                                                  |        |          |          |          |          |          |
| Total with Weighting                                             |        | 0        | -8       | -14      | 0        | -10      |

Additionally, we applied quantitative analysis and ADAMS simulations to the preliminary designs prior to selecting the final configuration. Rather than directly comparing power output, we normalized the models by calculating the time required to move from Target 1 to Target 4, setting the motor's maximum power output to 1W for consistent comparison.

**Table 2.2** Quantitative specifications of each design concept. This data influenced the Pugh chart rankings and was considered when we decided on manufacturability and potential design aspects.

| Option | Time Taken to move from Target 1 to Target 4(s/W) | Input Length (in) | Coupler Length (in) | Follower (in) | Max Transmission Angle Deviation (Degrees) |
|--------|---------------------------------------------------|-------------------|---------------------|---------------|--------------------------------------------|
| 1      | 0.39                                              | 7.3               | 2.2                 | 7.75          | 47.99                                      |
| 2      | 0.46                                              | 6.49              | 2.03                | 6.45          | 48.06                                      |
| 3      | 1.13                                              | 7.02              | 3.01                | 8.125         | 44.2                                       |
| 4      | 0.33                                              | 6.94              | 1.63                | 7.6           | 49.20                                      |
| 5      | 0.43                                              | 6.7               | 3.09                | 7.59          | 21                                         |

Overall, Option 1 tied with Option 4 the highest score on our Pugh chart and performed best on the metrics we prioritized. We ended up choosing Option 1 Although it scored lower in the volume and, therefore, weight category, we addressed this by specifying lightweight materials in the design phase, as detailed in Section 3. This approach results in a mechanism with reduced mass and, correspondingly, lower rotational inertia, while also maintaining a competitive advantage in incident light intensity on the target, thanks to its superior proximity to the target.

### Section 3: Final Linkage Design and Analysis

For the final design, provide tables of :

1. Final link lengths,
2. Transmission angle deviation for all linkage positions
3. Ground pivot locations relative to the bottom left corner of the setup area

Based on our decision using the Pugh chart and quantitative metrics, we finalized the design as option 1 and began optimizing it to increase performance and reduce weight. **Table 1**, **Table 2**, and **Table 3** provide in-depth specifications of our final concept.

**Table 3.1** Angular deviations of the leftmost and rightmost links of the final concept. Given the angle deviation data, we chose the leftmost link as the input link because its maximum deviation was lower than that of the rightmost link ( $43.96^\circ$  vs  $47.99^\circ$ ), with implications for the mechanism's mechanical advantage.

|                                    |                            | Target #1 | Target #2 | Target #3 | Target #4 | Average<br>(please calculate) |
|------------------------------------|----------------------------|-----------|-----------|-----------|-----------|-------------------------------|
| <i>leftmost link<br/>is input</i>  | Angle ( $\mu$ , degrees)   | 69.87     | 81.84     | 94.75     | 133.96    | 95.105                        |
|                                    | Deviation ( $ 90 - \mu $ ) | 20.13     | 8.16      | 4.75      | 43.96     | 19.25                         |
| <i>rightmost link<br/>is input</i> | Angle ( $\mu$ , degrees)   | 102.19    | 91.86     | 80.05     | 42.01     | 79.03                         |
|                                    | Deviation $ 90 - \mu $     | 12.19     | 1.86      | 9.95      | 47.99     | 18.00                         |

**Table 3.2** Input, Coupler, and Follower link lengths.

| Link     | Length (in) |
|----------|-------------|
| Input    | 7.30        |
| Coupler  | 2.20        |
| Follower | 7.75        |

**Table 3.3** Ground pivot locations relative to the bottom left corner of the setup area.

|                 | Distance from Bottom Left Corner (in) |
|-----------------|---------------------------------------|
| Leftmost Pivot  | 7.58                                  |
| Rightmost pivot | 8.73                                  |

### Alignment with Project Specifications

For the final linkage design, we are, as a group, constrained to have a transmission angle ( $\mu$ ) of between  $30^\circ$  and  $150^\circ$  when moving between the 4 different positions. Additionally, the transmission angle deviation which is defined by  $|\mu - 90^\circ|$  must be  $60^\circ$  at most. Our design for the final linkage design includes a maximum transmission angle of  $133.96^\circ$  and a minimum transmission angle of  $69.87^\circ$  which both fit within the accepted range for transmission angles. Additionally, the maximum transmission angle deviation was  $43.96^\circ$  which also adheres to the maximum requirement of  $60^\circ$ . This comparison is all shown below in **Table 3.4**.

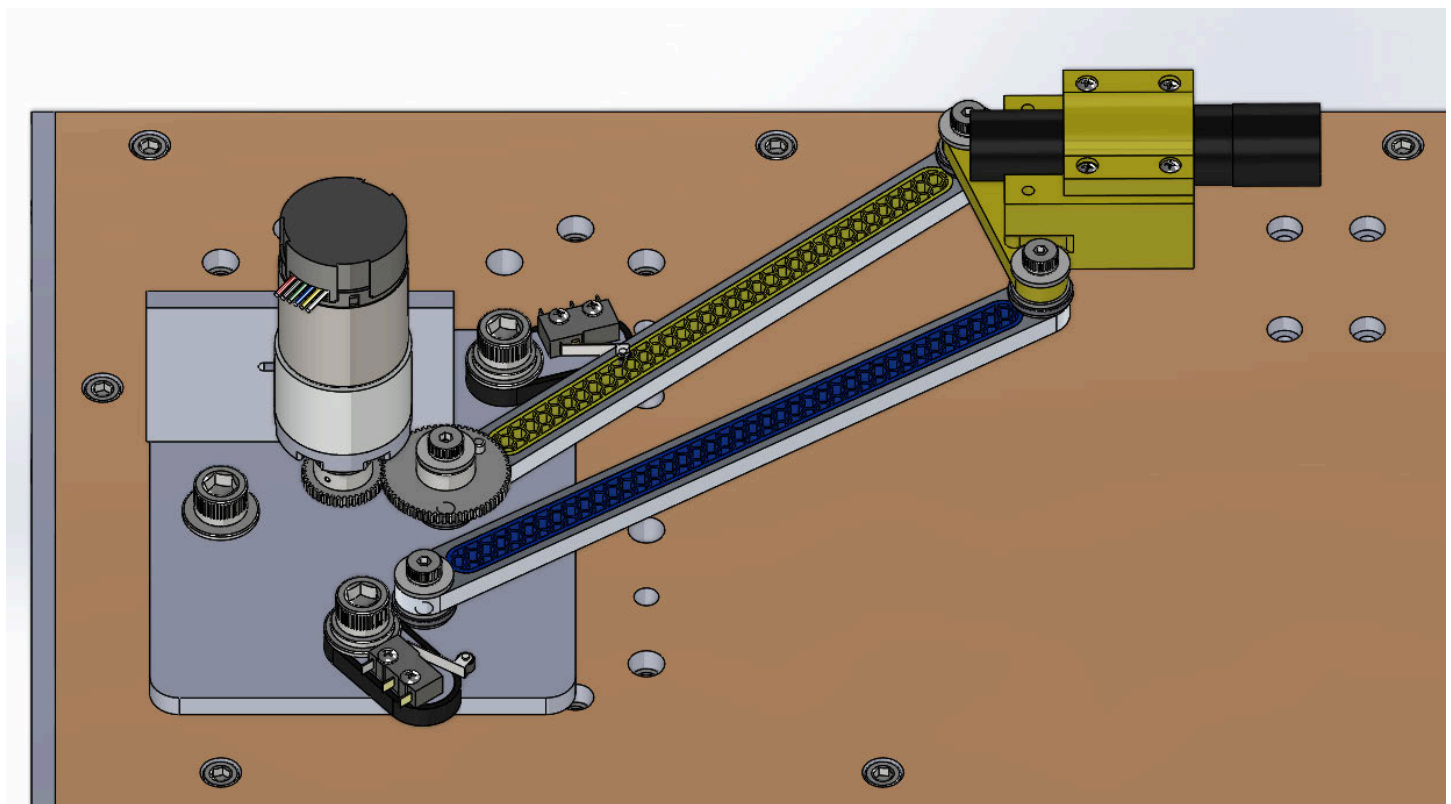
**Table 3.4.** Comparison of the project specification and our group's chosen linkage design's maximum transmission angle and deviation.



|                                         | Project Specification | Chosen Linkage Design | Adherence? |
|-----------------------------------------|-----------------------|-----------------------|------------|
| Maximum Transmission Angle ( $\mu$ )    | 150°                  | 133.96°               | Yes        |
| Maximum Deviation<br>$ \mu - 90^\circ $ | 60°                   | 43.96°                | Yes        |

### Final CAD Model

Included below is the final linkage design. The design is a culmination of the decisions outlined above in the Pugh chart and in our team's general discussion. The following sections discuss other aspects of the design, including the hardstops, material selection, and the adjustability of the linkage system as a whole.



**Figure 3.1.** The linkage across the four targets. Positions 1 and 4 are against the hardstops at each end of the range of motion

### Discussion of Design

**Hard Stops** - The design of the hard stops was driven by the priority of adjustability in the linkage, as well as by the presence of tolerances and imperfections. The hardstops consist of 1" diameter aluminum cylinders with an off-center hole. This off-center hole allows for the hardstop to be rotated and provides different distances from its mounting point to the contact point with the links. The cylinders will be rotated to account for any

imperfections in the manufacturing process or during installation of the linkage system on the playing field. This design should also accommodate quick, easy adjustments to improve testing and calibration efficiency.

The hard stops, baseplate, and linkage components are manufactured from  $\frac{1}{4}$ -inch aluminum stock. This material selection was influenced by both its ready availability and its capacity to provide sufficient rigidity during link–hard stop interactions at positions 1 and 4. For the inserts within the input and follower links, as well as the coupler and flashlight mount, PLA was chosen via 3D printing. Incorporating these 3D-printed PLA inserts significantly enhances the structural rigidity of the links while minimizing overall system mass. These inserts are engineered to mitigate link deformation upon high-velocity impacts with the hard stops during dynamic testing. The resultant mass reduction permits operation at increased angular velocities for a given power input. Furthermore, utilizing a fully 3D-printed coupler and flashlight mount reduces the applied load on both links and joints, thereby lowering frictional losses throughout the mechanism.

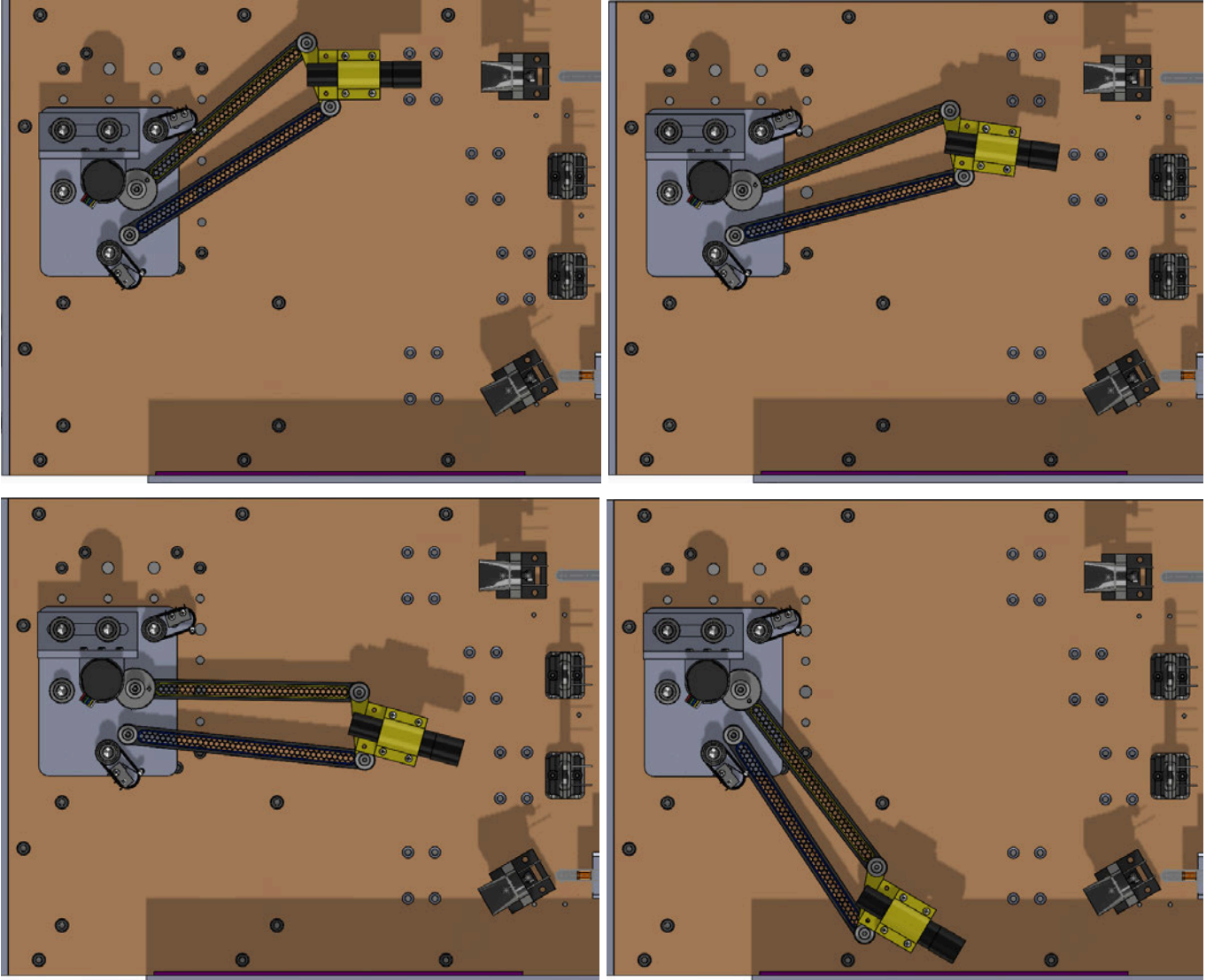
To optimize the link design for minimum weight, the thicknesses of the 3D-printed insert and aluminum walls were determined through conservative engineering calculations. The beam was modeled as a rigid body, and preliminary simulation data from our ADAMS model enabled the calculation of impact velocity at the contact point with the hard stop using rigid body dynamics. With this velocity, the corresponding impulse and resultant force on the beam were determined. These forces were then used as input for a detailed fatigue analysis and yield check, targeting a minimum service life of 100,000 cycles. The final design specified a total wall thickness of 0.205 inches. Multiple safety factors were incorporated throughout the analysis, including conservative assumptions for velocity, reduced second moment of area, point load approximations, and an elevated cycle count to ensure our design constraints are highly conservative.

The choice to use pla in the inserts for the links, coupler, and flashlight mount resulted in a greatly reduced weight. With the other links that are made of aluminum 6061 and joint assembly mostly composed of 841 bearing bronze, the total weight of the assembly came out to be an estimated 0.35 lb (with the infill density for the PLA parts assumed to be 25%).

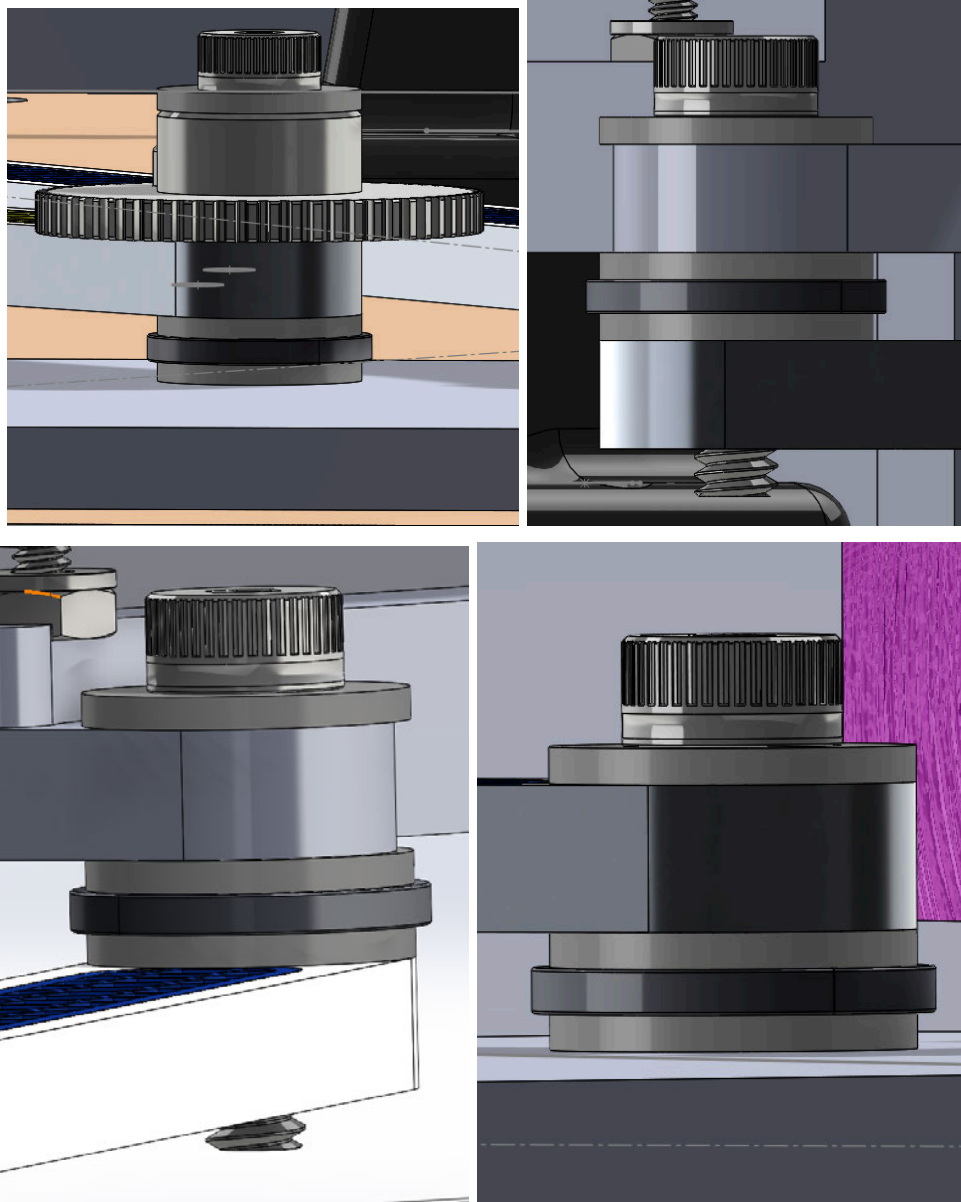
### Quality of Joint Design

3 of the 4 joints are of the exact same design, with the one exception being the joint connecting the input link to the baseplate, as that joint includes the gear attached to the input link. The majority of the joints use a  $\frac{1}{2}$ " long x  $\frac{1}{4}$ " diameter shoulder screw with #10-24 thread. The shoulder is made up of a needle thrust bearing surrounded by washers, a  $\frac{1}{4}$ " long x  $\frac{1}{4}$ " ID x  $\frac{5}{16}$ " OD bronze bushing that is pressed into the moving link, and another washer sitting between the moving link and the head of the shoulder screw. The fourth joint is made up of a similar design but with our output gear sitting on top of the moving link. This gear also has a bushing pressed into it. In this case, we are using a  $\frac{7}{8}$ " long shoulder screw to accommodate the gear.

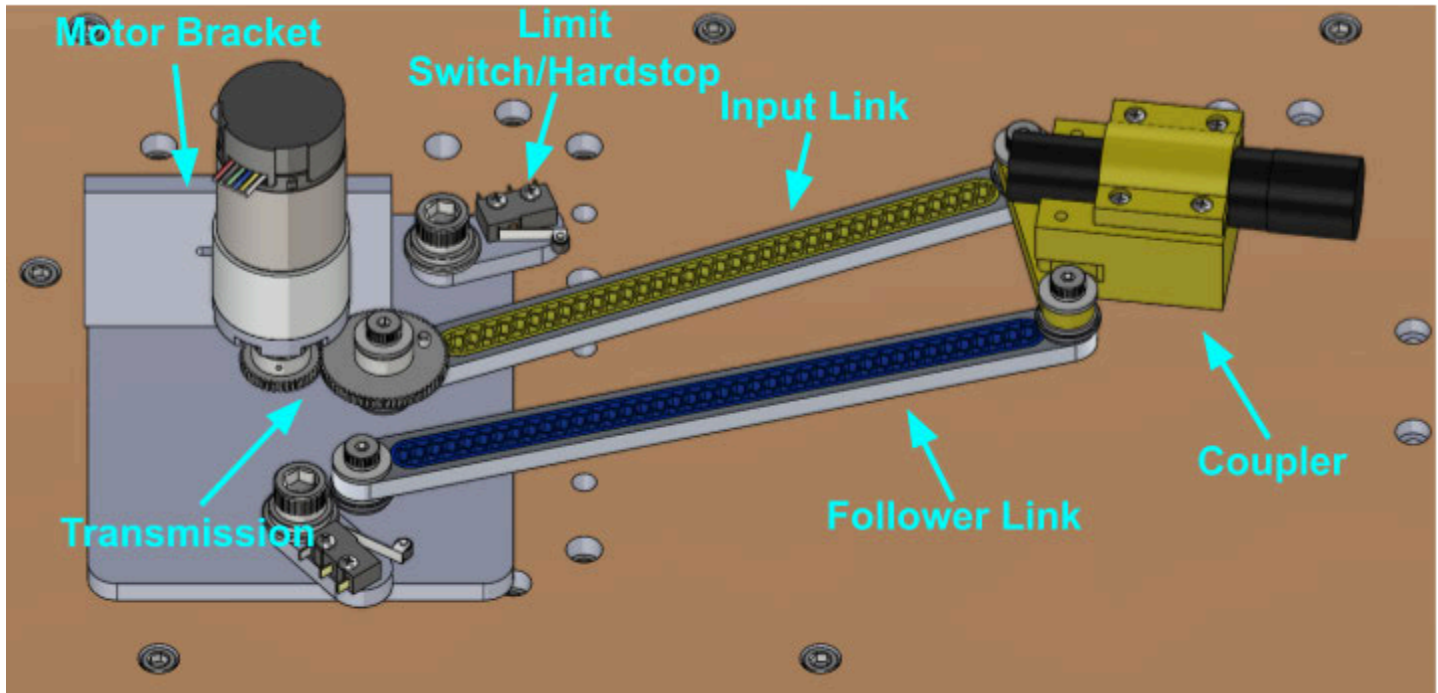




**Figure 3.2.** The linkage across the four targets. Positions 1 and 4 are against the hardstops at each end of the range of motion



**Figure 3.3.** From top left to bottom right, Input Link Joint, Coupler-Input Joint, Coupler-Follower Joint, and Follower-Ground Joint



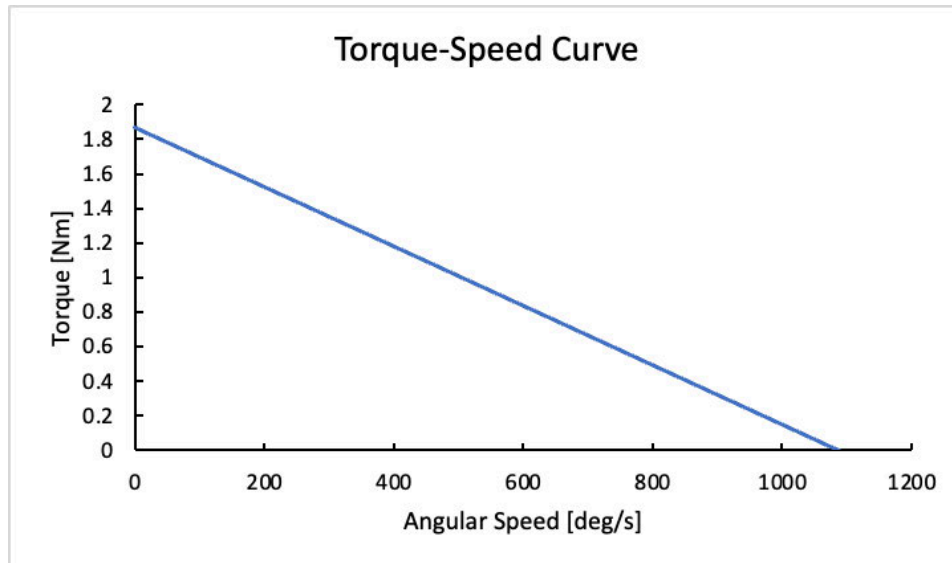
**Figure 3.4.** Perspective view of our full linkage design including the baseplate, transmission, hard stops, limit switches, and all fasteners.

### ADAMS Modeling Results

The ADAMS model was constructed using only the main components of the linkage, omitting the joint assemblies, fasteners, and other miscellaneous hardware components. This was done to drastically simplify the modeling process. The model included the input link, coupler link, follower link, flashlight, and flashlight bracket components. The mass of the links was determined using Solidworks, while the 3D printed parts were put into a slicing program, which then determined the final weight of the parts. While this model is the basis for our expectations for our linkage performance, the simplifications we made in modeling the system will certainly reduce the actual speed of the linkage.

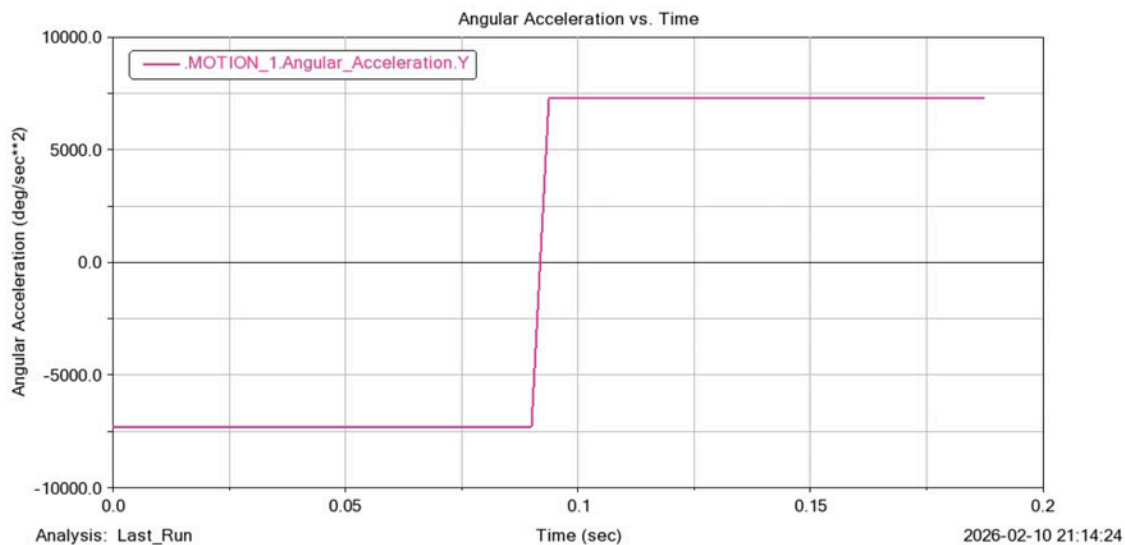
The motor-transmission combination used in our linkage, when run at 10 volts was found to have a stall torque of 1.87 Nm, a no-Load speed of 1088 deg/s, and a maximum power output of 8.8 watts. These figures were found under the assumption that the chosen transmission would be 100% efficient. During the modeling of the linkage, we ensured that every operating condition fell under the steady state operating conditions outlined by the calculated Torque-Speed curve, while also never exceeding the calculated maximum power.

The results of the ADAMS modeling yielded a minimum projected time to drive the linkage from position 1 to position 4 of 0.1875 seconds. This is achieved while remaining well within the bounds of the motor limits. The maximum torque required for this motion is 0.55 Nm, while the maximum angular speed is 684 degrees/second, and the maximum required power output of the motor is 6.3 watts. This number, while conservative is likely smaller than the actual achievable speed of the linkage, since the model did not include several components that would increase the torque demanded, as well as introduce power losses in the linkage.



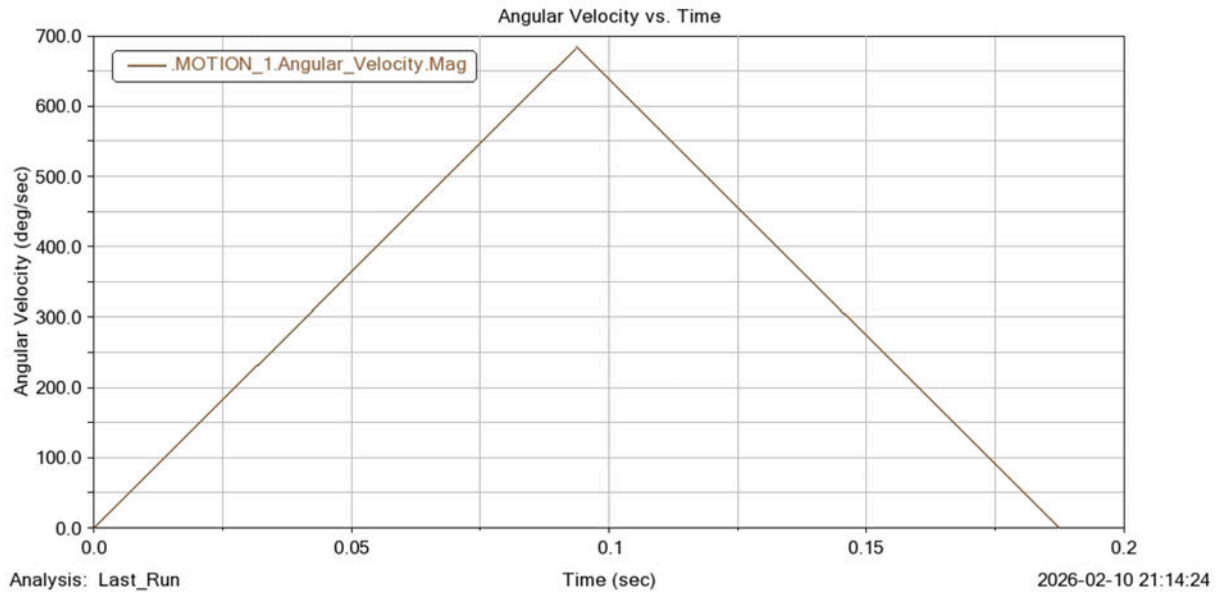
**Figure 3.5.** The characteristic Torque-Speed curve calculated for the motor-transmission setup chosen. The motor should be able to perform at any operating condition below the line shown.

The acceleration profile was chosen to be a step function of constant acceleration, followed by equal deceleration, this was chosen to minimize the time that the linkage takes to rotate between the first and the fourth positions. The acceleration profile is shown below in **Figure 3.6**.

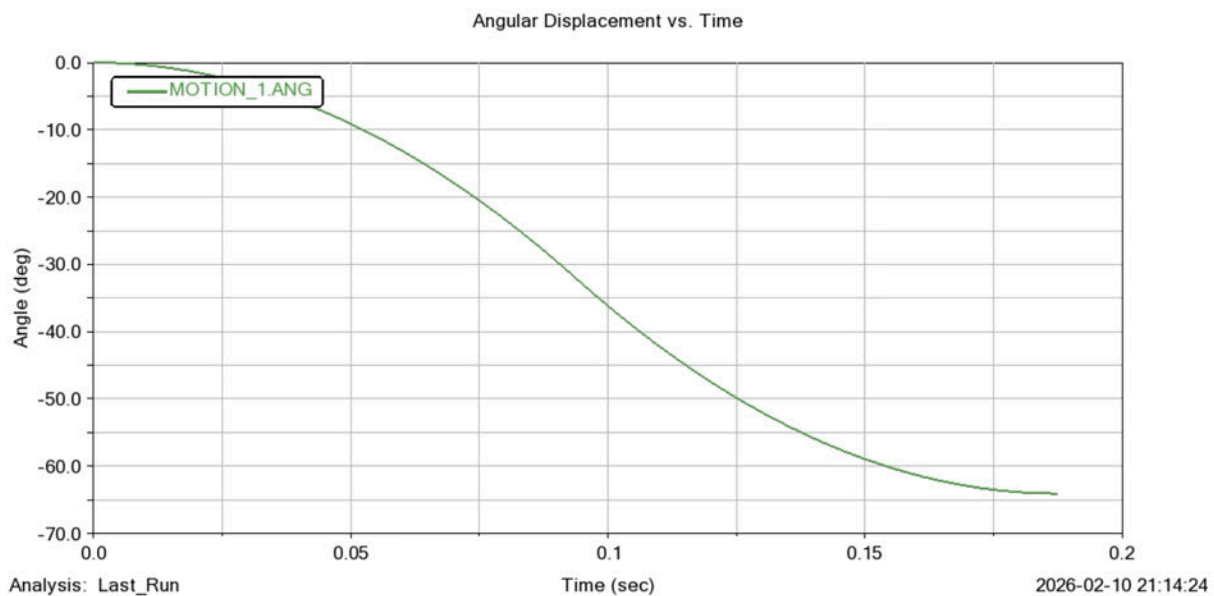


**Figure 3.6.** The acceleration profile of the input is a step function, operating at -7292 degrees/second and 7292 degrees/second.

This acceleration profile yields a velocity profile that peaks at  $\frac{1}{2}$  the total transition time. While iterating through different acceleration values this profile had to be examined to determine that the velocity did not exceed the limits of the provided motor while connected to our transmission. The velocity and displacement profiles are shown below in **Figures 3.7** and **3.8**

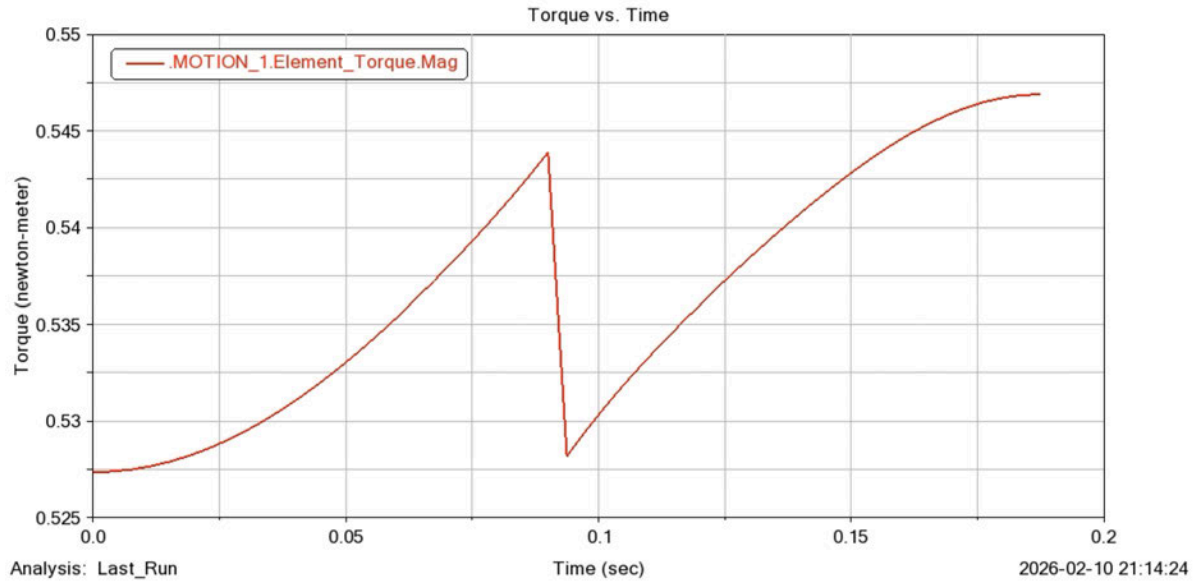


**Figure 3.7.** The maximum angular speed of the motor determined in ADAMS was 683.62 degrees/second. This was substantially lower than the maximum possible speed for the motor.

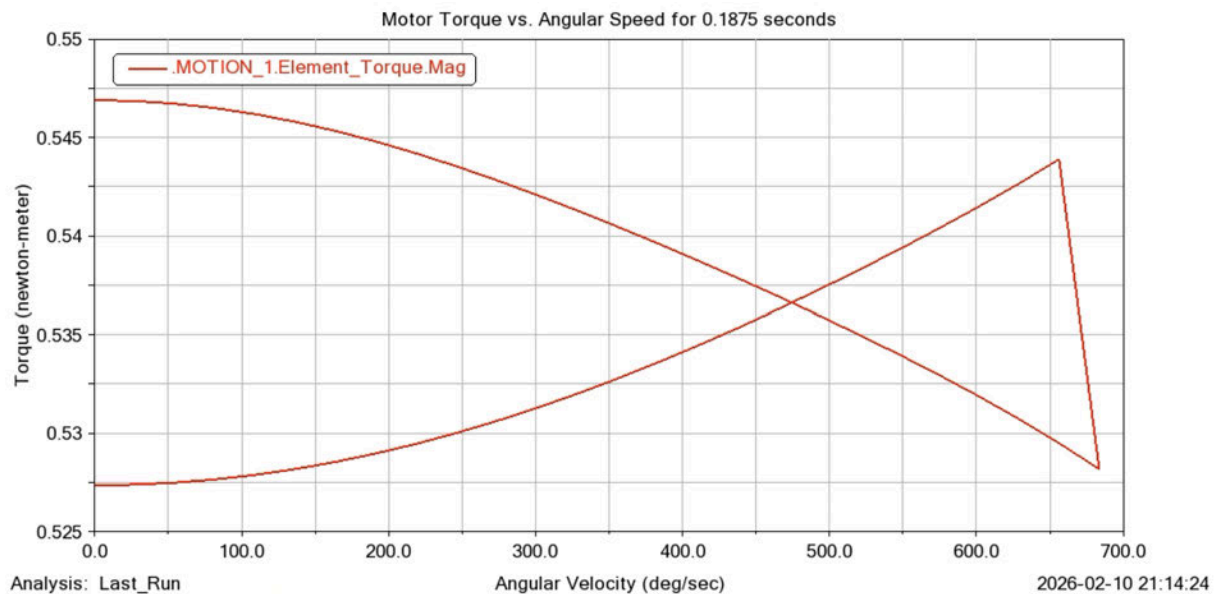


**Figure 3.8.** The velocity profile created by the motor input produces a smooth curve from position 1 to position 4.

The torque required to drive the linkage followed a much rougher and varied profile than the motion characteristics. This is primarily due to the instantaneous change between acceleration values, thus requiring a spike in torque demanded from the motor. In practice, this spike would be expected to shrink, and thus the acceleration profile would not be an exact step function. In addition, a plot comparing the torque and the angular speed of the motor-transmission system was constructed to ensure that the demanded torque at all angular speeds could be supplied by the motor.



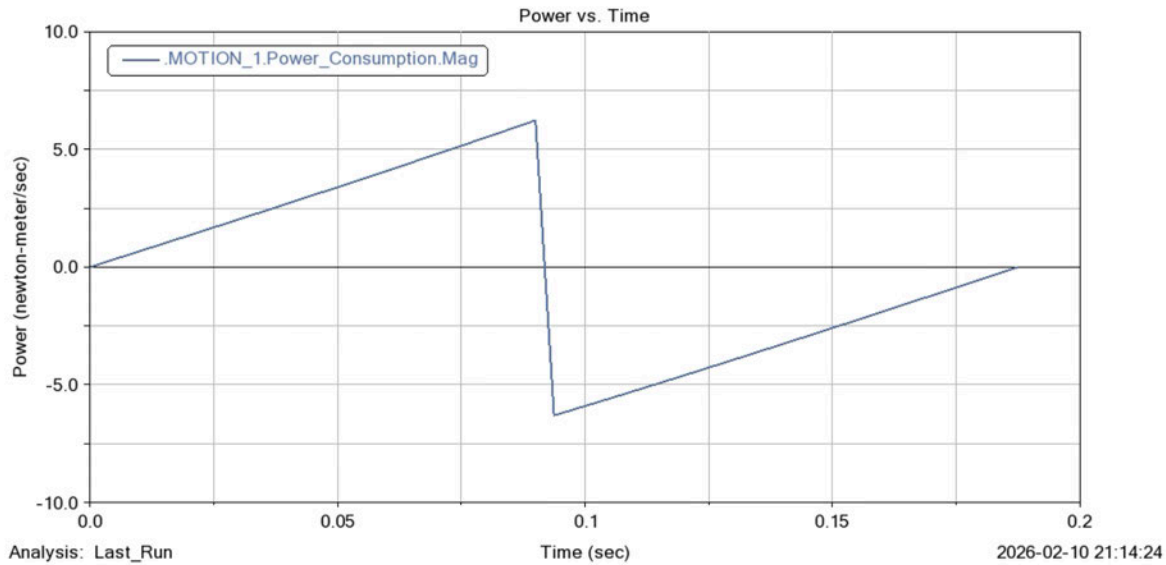
**Figure 3.9.** The torque of the input follows a unique profile, with two maxima, and two minima. The maximum torque required is 0.55 Nm, which is less than  $\frac{1}{3}$  the calculated stall torque of the motor-transmission input.



**Figure 3.10.** The Torque-Speed curve shows a clear maximum power demand around where the input is spinning at its maximum speed during the motion. Even with this peak demand, the entire curve is well within the bounds of the motor performance curve. At the peak power demand of the curve, the input is required to rotate at 656 degrees/second, with a torque of 0.54 Nm, while the maximum torque deliverable at this speed is calculated to be 0.74 Nm.

The power profile for the ADAMS model is relatively simple, with a steady increase in power demand achieving a maximum value immediately before deceleration. At no point does the power demand exceed the maximum power limit.





**Figure 3.11.** The maximum power required to drive the system with the given acceleration profile is 6.3 watts, which is substantially less than the calculated maximum power of 8.8 watts.

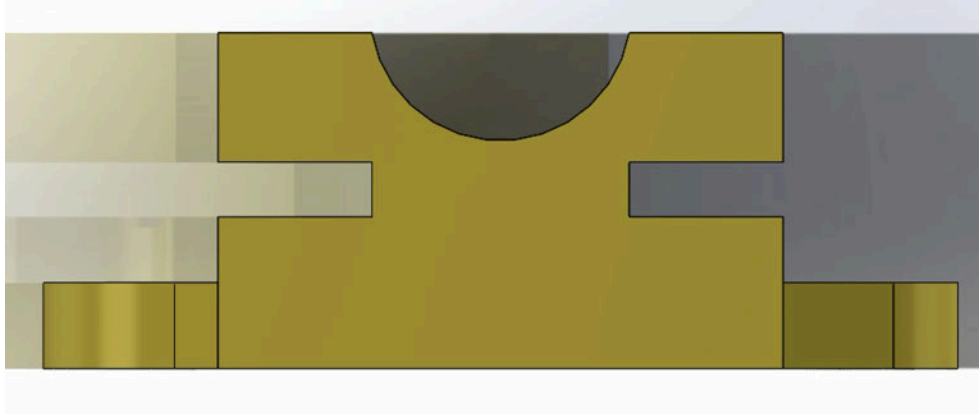
While the actual performance of the linkage is expected not to adhere exactly to the performance observed in the ADAMS model due to extra mass and sources of power loss, the time that the mechanism takes to move between the outermost positions is not expected to change much. This is due to the fact that the motor is not expected to be run at the limits of its performance capabilities.

#### Section 4: Motion Generator Revision (0.25-0.5 pg):

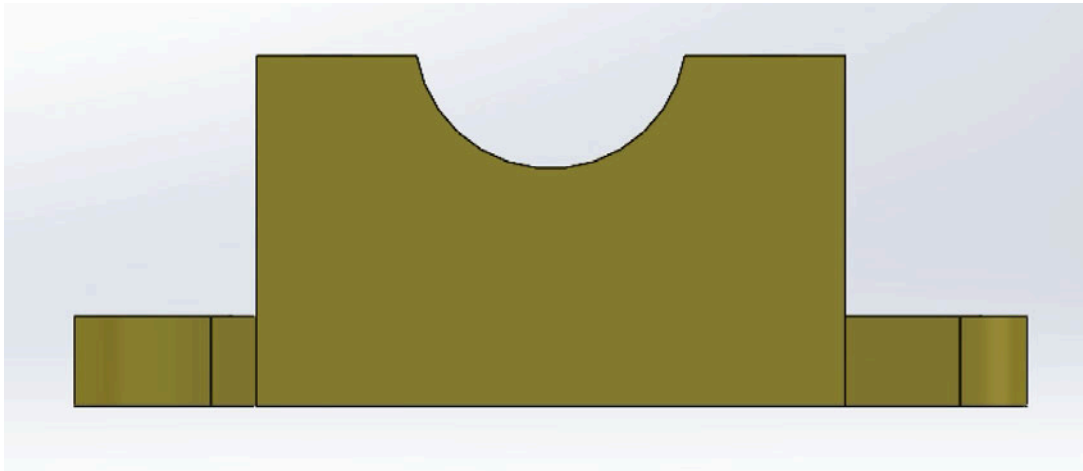
Since building and assembling a working model of our linkage, the team has noticed a few potential problems, and the designs have been subsequently modified to address them. This section summarizes these changes in detail.

The coupler was subject to two distinct issues, which were resolved by making a simple design change. The first issue was how difficult it was to properly set the 4-40 nuts into the coupler while simultaneously threading the screws. This made fastening the flashlight to our linkage take longer and required both a wrench and a screwdriver. In addition, the coupler was not strong enough to both apply sufficient clamping force to the flashlight and withstand the forces involved in the linkage's motion.

These problems were solved by simply removing the overhanging section of the coupler and instead making a fully enclosed space for the nuts to be set into. By removing this overhang, the coupler strength increased substantially, while only marginally increasing the mechanism's weight.

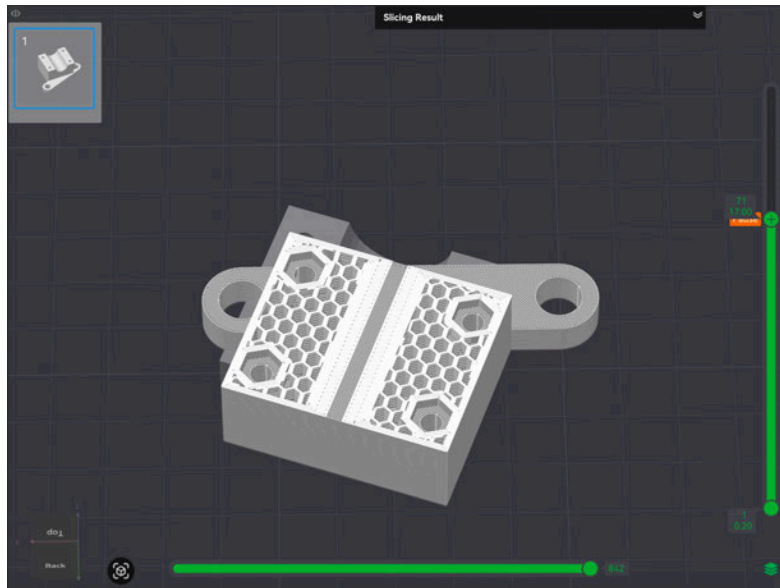


**Figure 4.1.** The original coupler design featured an overhang, which allowed for 4-40 nuts to be placed in a small recess, and provide the threading by which the clamping screws could be affixed to the coupler.



**Figure 4.2.** The updated coupler design does not have an overhang, which greatly improves the strength and rigidity of the link while experiencing load.

The nuts were set into the coupler by designing a press-fit hole to house them within the coupler body. No longer need to perfectly place the nuts; instead, they can be pressed in at a specific layer line by automatically pausing the print. After inserting the nuts, a thin layer of glue can be applied to ensure the subsequent layers adhere properly to the hardware surface. This change made fastening the flashlight to the linkage significantly faster and removed the need for a wrench to keep the nuts from rotating with the screw.



**Figure 4.3.** The slicing page for the coupler print. As shown in the image, the top of the nut housing corresponded to the 71st print layer. By pausing the print, the nuts could be inserted before the next print layer covered the housing.

The second design flaw identified in Design Review 1 was the method used to mount the hard stops to the baseplate. The original design used mounting screws to hold the hardstop fixed relative to the baseplate and to secure the baseplate to the playing field. This means that every time the mechanism is mounted for testing, the hard stops have to be readjusted. As a result, repeatability of the hardstop location was compromised, and additional time would be required to complete the mounting procedure on the day of the competition.

To fix this, the baseplate mounting holes were changed from clearance holes for a  $\frac{3}{8}$ " screw to tapped holes for a  $\frac{1}{4}$ -20 mounting screw. In addition, the mounting holes for the hardstops were changed to clearance holes for a  $\frac{1}{4}$ " screw. Neither hole was moved because the new  $\frac{1}{4}$ -20 mounting screws fit inside the  $\frac{3}{8}$ -16 tapped holes of the playing field with sufficient clearance. By optimizing this design, our team can devote more time and attention to the more technical components of testing, rather than wasting time setting up the mechanism for each lab.

## Section 5: Energy Conversion Introduction

### Why is a transmission needed?

The purpose of our transmission is to effectively transfer power from the motor to the input link. A transmission serves as a critical intermediary between the motor and the load, compensating for the motor's inability to directly satisfy the torque and speed demands of the linkage specified in our ADAMS model.

### What are the effects of transmissions on speed, torque, and/or other important quantities for the project?

Transmissions are used to change force/torque, direction & speed between an input and output. For our project, a transmission is used to optimally transfer the load from point A to point B. In our case, we are moving the linkage from the first target to the fourth. Transmission ratios are governed by the following equation, accounting for transmission efficiency:

$$N = \frac{\omega_m}{\omega_L} = \frac{T_L}{\eta^* T_m}$$

Where  $N$ ,  $\omega_m$ ,  $\omega_L$ ,  $T_L$ ,  $T_m$ ,  $\eta$  are the Transmission Ratio, Angular Speed of the input, Angular Speed of the Output, Output Torque, Input Torque, and Transmission Efficiency respectively. Finding the optimal transmission ratio optimizes the torque requirements of the load and the rotational speed to achieve the fastest time between targets 1 and 4. This is also known as load-inertia matching meaning the inertias of the motor reflected on the same shaft are matched. The concept of Load-Inertia Matching is further discussed in Section 6.

### What are the limits of any general transmission design?

Theoretically, an ideal transmission system has no losses associated with it. However, in practice, a transmission experiences losses due to friction, slipping, and heating, all of which can be reduced using effective lubrication.

### What are the design limits for the transmissions in this project?

One of the important considerations we had to make was cost. Given our limited budget, we chose to use Nylon gears from McMaster Carr. While they are rated at torque values much higher than those reported by our ADAMS model, we still expect greater degradation over time and must be more careful during testing. Specific to our case, because we chose spur gears, in addition to the general limits of transmission design, there are constraints due to backlash, which reduces positional accuracy. Altogether, these project-specific considerations should not significantly impact project performance, provided we tune the system carefully.

## Section 6: Transmission Ratio and Type Determination

We determined our transmission ratio by considering inertia matching and light beam resolution analysis, and selecting the higher of the two gear ratios.

### Inertia Matching

Since the links are moving during operation, the inertia around the motor is constantly changing. Since inertia is most important when changing direction and starting motion, we considered only the moment of inertia at the 4 predetermined positions on the playing field.

The Load-Inertia Matching equation goes as follows:

$$N_{opt} = \sqrt{\frac{J_L}{J_m}}$$

Where  $N_{opt}$ ,  $J_L$  &  $J_m$  are the Optimal Transmission Ratio, lumped load-side gear inertia & lumped motor side gear inertia.

To estimate the lumped load inertia, the mass and centre of mass of each component were found on SolidWorks. Next, the distances between the components and their axis of rotation were considered at 4

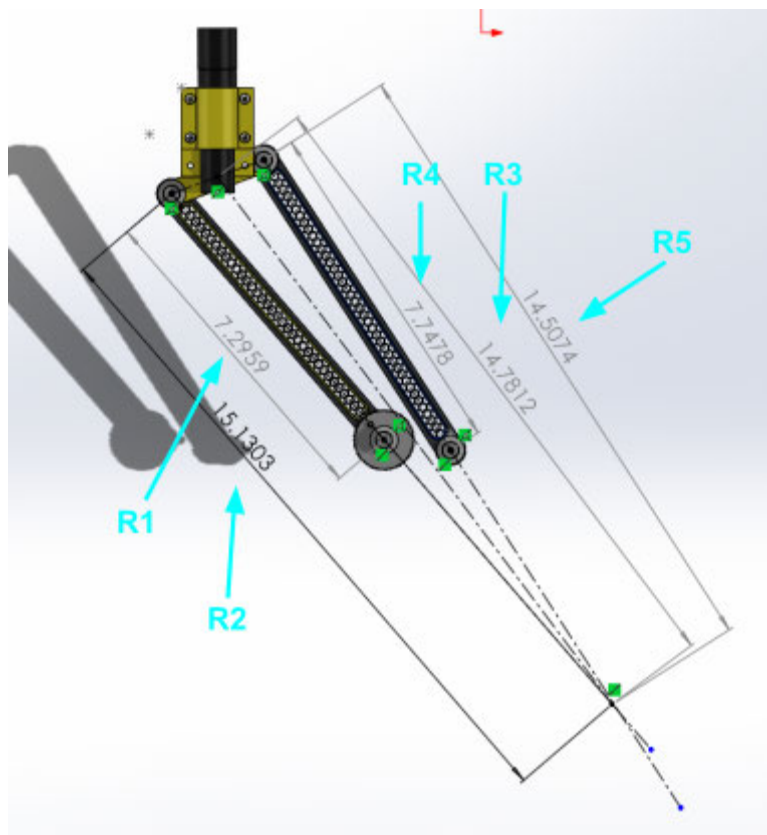
positions. Altogether, the total mass moment of inertia was calculated using the following equation, which stems from the Parallel Axis Theorem:

$$\frac{1}{2}I_{total}\omega_{input}^2 = \frac{1}{2}I_{input}\omega_{input}^2 + \frac{1}{2}I_{coupler}\omega_{coupler}^2 + \frac{1}{2}I_{output}\omega_{output}^2$$

Of the four positions tested, the maximum mass moment of inertia was selected as a conservative estimate for calculating the optimal transmission ratio. While intermediate positions between the four evaluated configurations were not considered, this simplification has little practical impact, since the mass moment of inertia most significantly affects startup torque, and the mechanism only starts and stops at the four positions tested.

Motor inertia was estimated by referencing the motor manufacturer's specifications. This value is inherently an approximation, as it is derived from idealized manufacturer data rather than our own empirical tests.

Combining the maximum load-side mass moment of inertia with the estimated motor-side inertia, a conservative optimal transmission ratio was calculated. This analysis yielded a maximum transmission ratio of 1.55.



**Figure 6.1.** Demonstration of how the R values were calculated for our mechanism. The linkage is shown in position one, and the demonstrated measurements are in inches compared to the calculated values in centimeters.

**Table 6.1.** Comparison of the project specification and our group's chosen linkage design's maximum transmission angle and deviation.

|         | Position 1 | Position 2 | Position 3 | Position 4 |
|---------|------------|------------|------------|------------|
| R1 (cm) | 18.52      | 18.52      | 18.52      | 18.52      |
| R2 (cm) | 37.31      | 51.17      | 69.64      | 52.99      |
| R3 (cm) | 37.36      | 50.56      | 70.35      | 57         |
| R4 (cm) | 19.67      | 19.67      | 19.67      | 19.67      |
| R5 (cm) | 40.62      | 53.21      | 72.49      | 57.32      |

### Light Beam Resolution Analysis

Ensuring sufficient sensor resolution to accurately position the light beam is a key design requirement. Since our final code uses the motor's angular position as an input, a larger transmission ratio improves positional precision at the cost of speed. To evaluate this tradeoff, we analyzed the light beam resolution using our SolidWorks model. Assuming a light beam thickness of 1 inch, we calculated the maximum and minimum angles between the bottom of the game board and the input link required for the light beam to cover half of the light intensity sensor. This process was repeated at every sensor on the game board, and the angular difference at each position was recorded. The smallest difference found was 4.58 degrees.

From there, we selected the minimum recommended desired count range of 30 counts — defined as the number of encoder counts (provided by the manufacturer as 64 counts per revolution with quadrature) that the linkage can move while still landing the light beam on the target. This yielded a desired transmission ratio of 1.22. The following equation was used to find the desired count range:

$$\text{Desired Count Range} = \left( \frac{64 \text{ counts}}{1 \text{ rev}} \right) \cdot 30 \cdot \left( \frac{\text{smallest difference [deg]}}{\frac{360 \text{ degrees}}{1 \text{ rev}}} \right) \cdot N$$

By combining inertia matching with light beam resolution analysis, we identified a range of transmission ratios that balance the competing demands of speed and positional precision. Since the ratio derived from light beam resolution analysis was lower than that from inertia matching, we selected the inertia matching ratio as the final value. A higher transmission ratio further improves light-beam resolution, making this a conservative and well-justified choice.

### Final Transmission Selection

When making the decision on the final transmission and transmission ratio, we considered all types of transmission types. **Table 6.2** is a Pugh chart of the different transmission types we considered.



**Table 6.2.** Pugh Chart comparing three transmission types — Belt Drive (baseline), Spur Gear System, and Chain Drive — where each alternative is scored “+1” if it outperforms the baseline and “-1” if it does not. Evaluation criteria were weighted toward cost, given our strict budget constraints, as well as precision and mechanical losses. Precision is critical to ensuring the concentrated light beam is accurately focused on each target, while minimizing losses ensures that motor output is directed toward driving the linkage rather than overcoming friction. Based on these criteria, the spur gear system emerged as the most suitable choice.

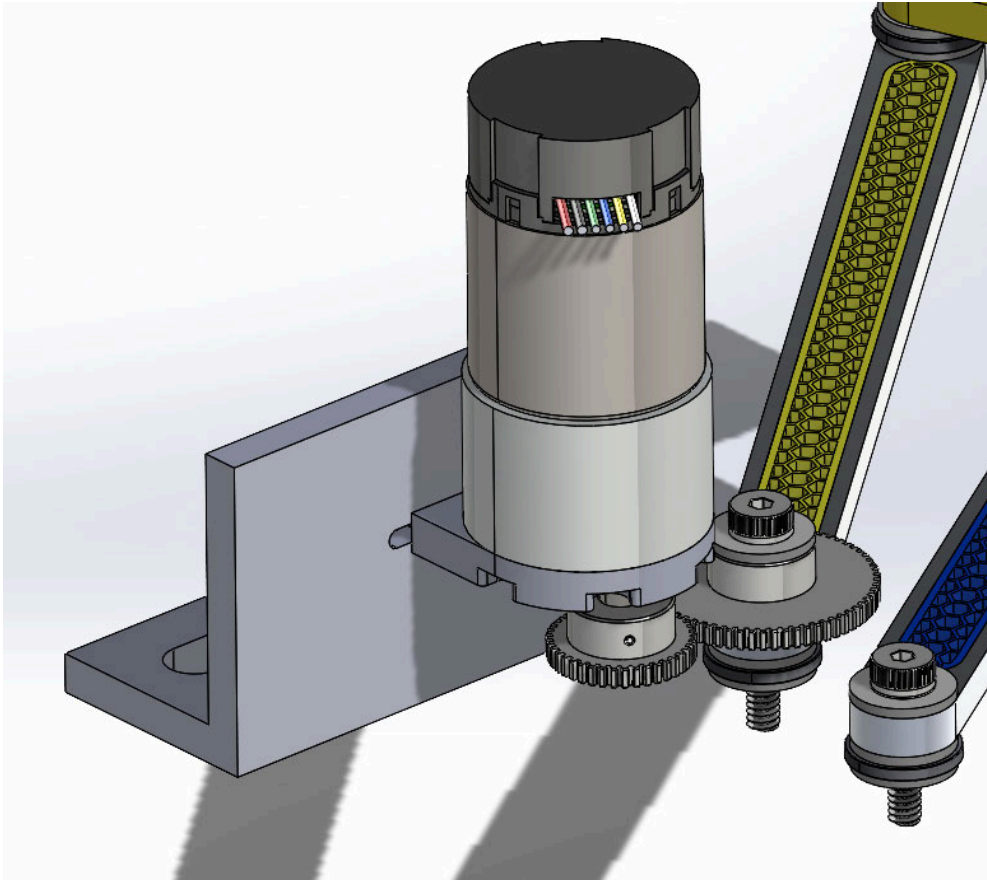
|                  | Cost     | Losses   | Noise    | Maintenance | Precision | TOTALS |
|------------------|----------|----------|----------|-------------|-----------|--------|
| <b>Weighting</b> | <b>3</b> | <b>2</b> | <b>1</b> | <b>2</b>    | <b>3</b>  |        |
| Belt Drive       | 0        | 0        | 0        | 0           | 0         | 0      |
| Spur Gears       | 1        | -1       | 1        | 1           | 1         | 6      |
| Chain Drives     | 1        | -1       | 1        | -1          | 1         | 4      |

Based on the Pugh chart analysis, spur gears were selected as the transmission type. To manage costs, nylon spur gears were chosen over metal ones — while metal gears would have been the ideal choice mechanically, nylon alternatives rated well above our calculated torque requirements using our ADAMS model offered a budget-friendly solution. Full part specifications are available in Appendix B. A transmission ratio of 1.5:1 was selected as the closest available option to the optimal ratio determined through inertia matching.

## Section 7: Final Transmission Design

For our final transmission design, we spent significant time deciding how best to attach the load-side and motor-side gears to their respective shafts. We were careful to make decisions that wouldn't compromise the structural integrity of the Nylon spur gears or wear down the teeth during testing. As mentioned before, due to budget constraints, we decided to use Nylon gears rated higher than the expected Torque and stresses from operating the mechanism. Refer to section 8 for further information on how this was calculated. Another consideration is that Nylon spur gears are much lighter than metal gears. Given that we are attaching the gear and dowel pin to the input link itself, the gear and dowel pin's weight are included in the mechanism's weight. The additional mass from the spur gear and dowel pin is only 5g of the mechanism's total weight. Altogether, this leads to a total mechanism weight of 165g.

We also spent considerable time designing our L-bracket that holds the motor, allowing adjustment of the transmission to ensure optimal meshing between the gears. We also considered adding the necessary adjustability in case we need to replace the gears or want to rotate the motor or linkage independently. Our L-bracket design utilizes a slot that allows us to move the motor independently of the L-bracket, and another slot to move the L-bracket independently of the base plate. In the event of gear failure, replacement gears available in the MEx50 room can be readily substituted, given that we factored in adjustability in our transmission design.



**Figure 7.1** shows the final transmission design, including the motor mount and torque-transferring joints. In this case, a dowel pin is attached to the input link, and a spring pin is attached to the motor shaft. Beyond gear-to-gear torque transmission, 1/16" spring pin and 1/8" steel dowel pin are used to transfer torque at two critical interfaces: between the motor shaft and the input gear, and between the output gear and the input link, as shown in the figure.

## Section 8: Torque Transfer Analysis

The final transmission design achieves a 1.5:1 gear ratio using two nylon gears with 64 and 40 teeth, respectively, to transfer torque from the motor to the linkage. A primary design concern was gear failure under high starting torque or during low-speed direction reversals — both conditions that place significant stress on nylon components. To verify structural integrity, the maximum allowable torque was evaluated using the Lewis bending equation:

$$T = \frac{D_p \sigma_F Y}{2P}$$

Of the two gears, the 40-tooth gear was the focus of the analysis due to its lower Lewis form factor ( $Y = 0.336$ ) and smaller pitch diameter ( $D_p = 0.833"$ ). The shared parameters (diametral pitch  $P = 48$ , face width  $F = \frac{1}{8}"$ , and tensile strength of 12,000 psi) yield a maximum allowable torque of 4.37 lb·in (0.49 N·m). The maximum input gear torque, derived from the ADAMS simulation and adjusted by the gear ratio, is 0.36 N·m, resulting in

a safety factor of 1.36. To ensure proper gear meshing during assembly, an adjustable L-bracket and motor mount are used to precisely align both gears, promoting even load distribution and efficient torque transfer.

Between the motor and the input gear, the pin is inserted through the motor shaft and contacts both sides of the gear to distribute torque evenly. The chosen dowel pin has a listed double-shear breaking stress,  $F_{ds}$ , of 2600 lbf. To find the maximum torque these pins can withstand before yielding, we multiplied the double-shear breaking stress by the motor shaft radius. With our shaft diameter of 6mm, the maximum torque we found was 307 lb-in (35 N-m), yielding a safety factor of 63, with the maximum torque at the input gear previously found to be 0.36 Nm. Additionally, the pin between the gear and the input link is in single shear, meaning it can withstand only half of the double-shear breaking stress of 1300 lbf. To convert this into torque, we multiplied this force times the distance between the center of the gear and the dowel pin, which is 0.46". This works out to 780 lb-in (88 N-m), giving us a safety factor of 162 relative to the maximum torque of our motor and transmission, 0.54, found in our ADAMS model.

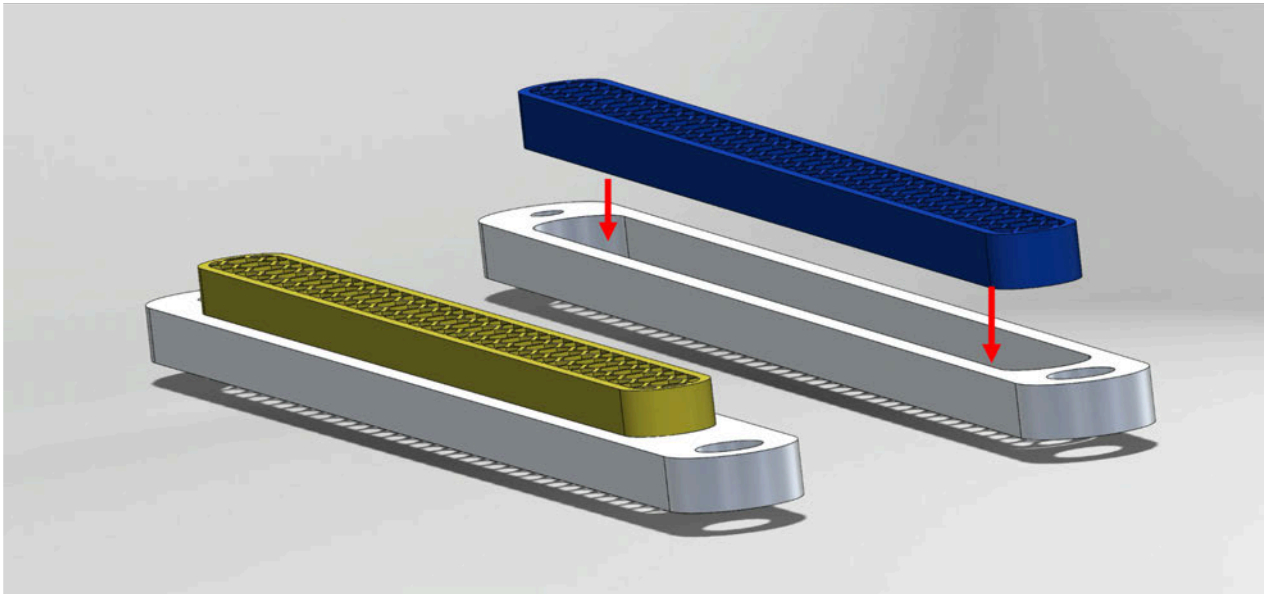
With the previously determined safety factors for the amount of torque transmitted between the gears and between the motor shaft and the input gear that dowel pins can transmit, our team is confident that our design will be able to effectively transfer high starting torques and low speed torque from reversing directions without failing due to yielding through shear or bending stress.

## Appendix A: Assembly Manual

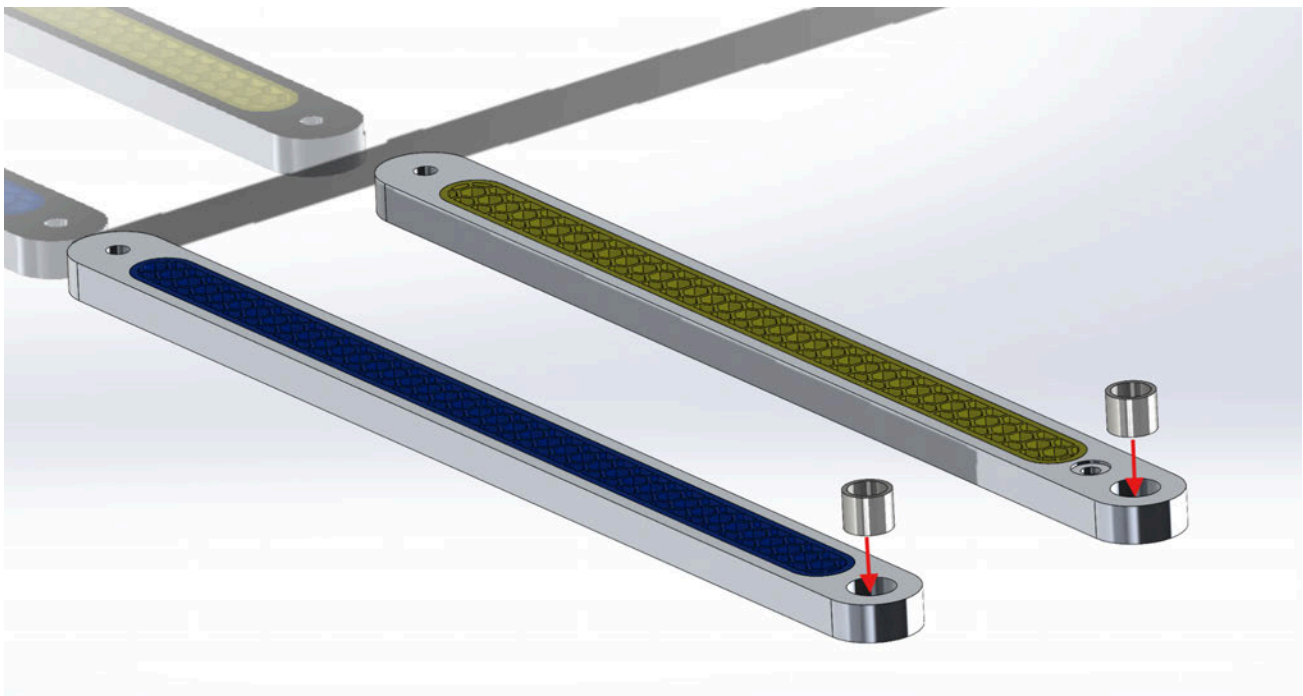
Append to original assembly plan as necessary

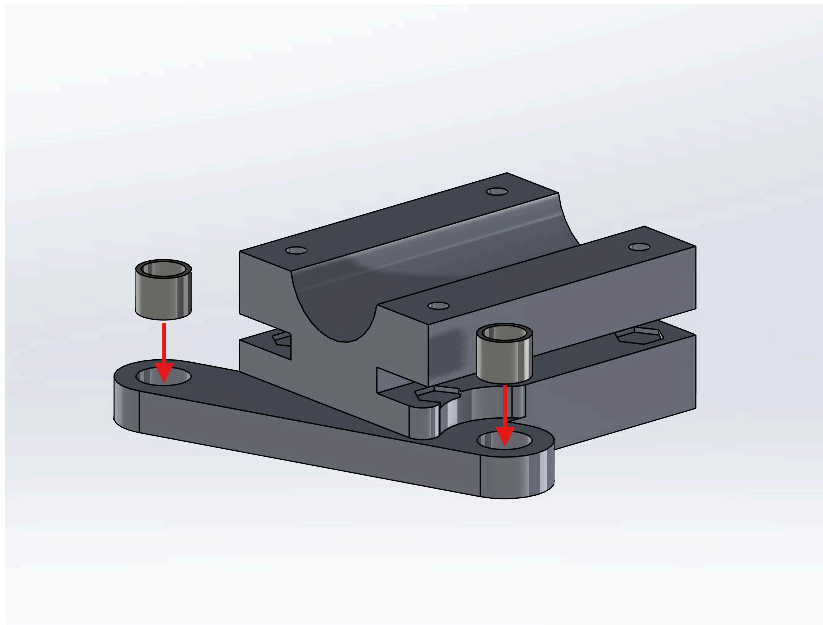
This section should resemble an IKEA instruction set for the transmission design. Break the entire assembly process into at least ten steps for linkage mechanical assembly (DR1), and at least ten steps for transmission and motor assembly (DR2). Provide a written description and picture for each step. Label and number the components in each picture.

1. Press-fit the 3d-printed inserts into Follower and Input Links

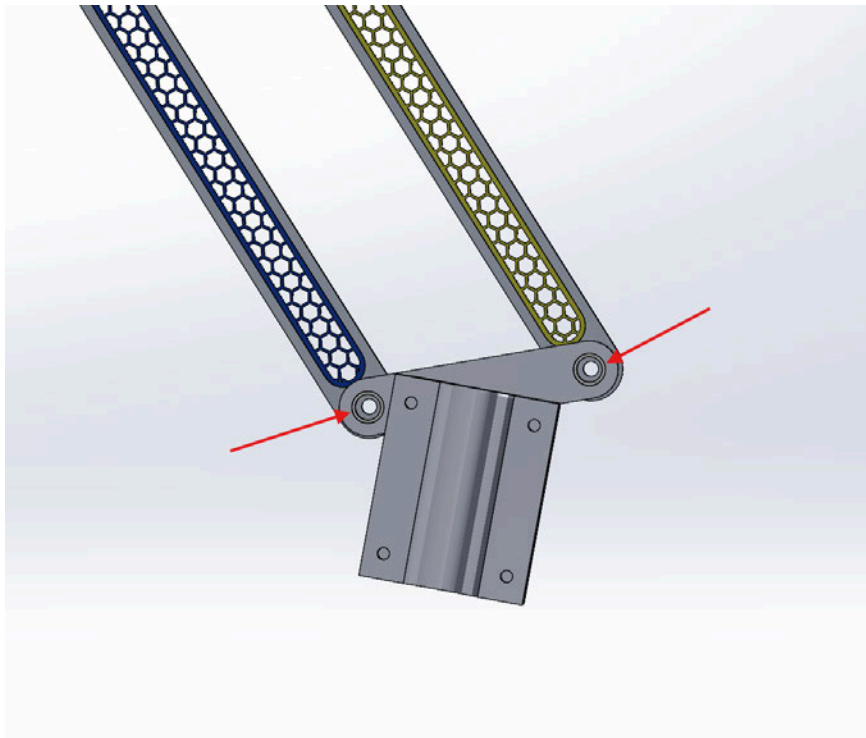


2. Install Bushings into the input and follower links as well as the coupler

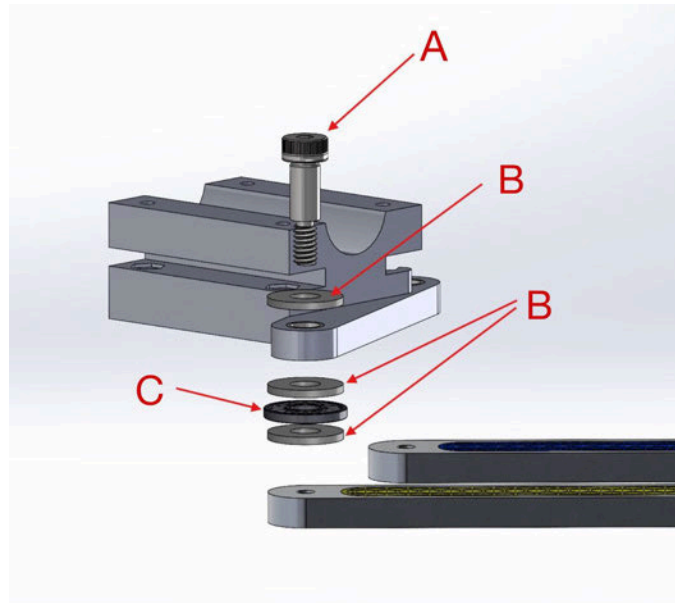




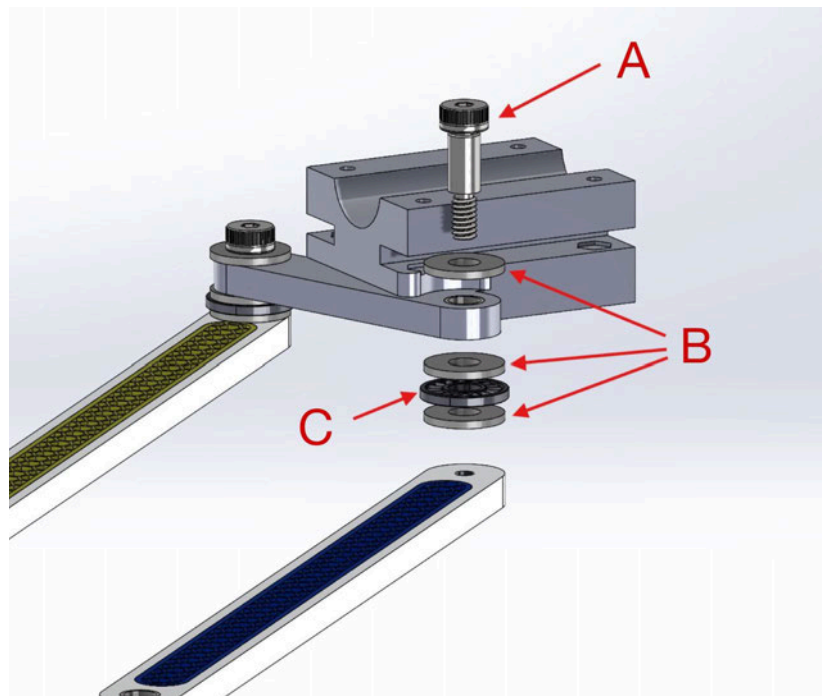
3. Align Follower & Input links with Coupler. Ensure this orientation is maintained as you add the joints.

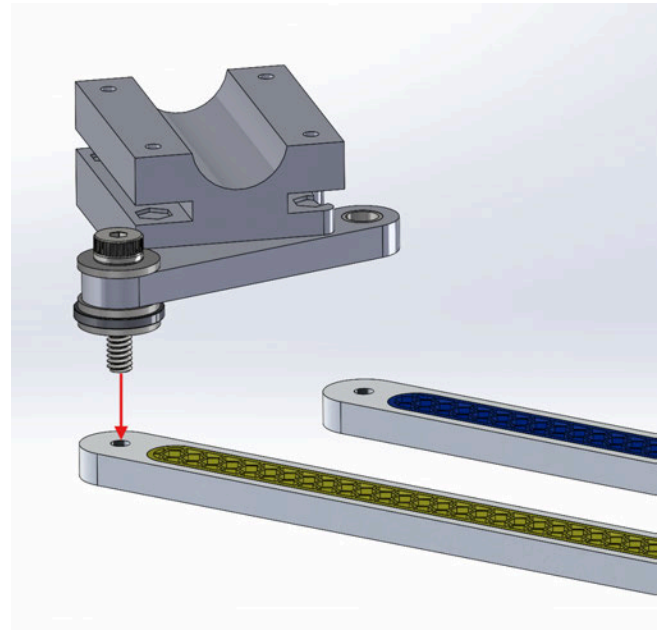
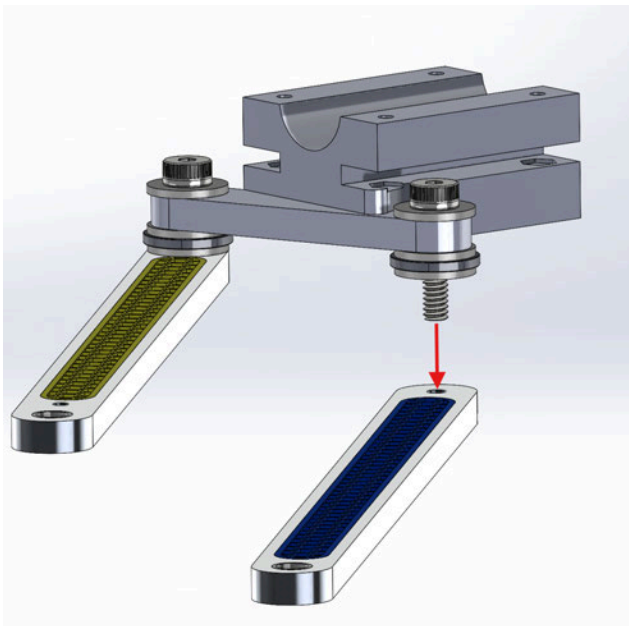


4. Finish joint assembly that mates Input link and coupler and then attach with the bolt  
A-  $\frac{1}{4}$ " shoulder diameter,  $\frac{1}{2}$ " shoulder length, 10-24 thread shoulder bolt  
B- Oil embedded-thrust bronze bearing  
C- Needle Thrust Bearing, Bore  $\frac{1}{4}$ "

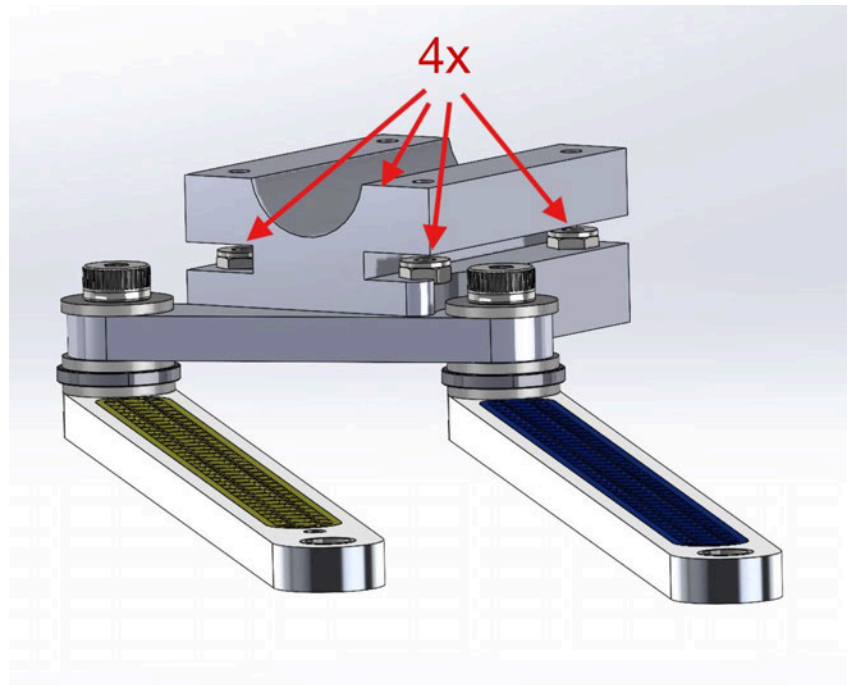
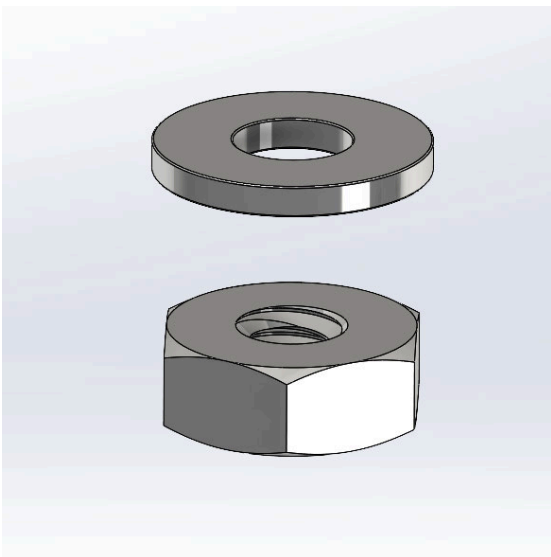


5. Finish Joint assembly that mates follower link and coupler and then attach with the bolt  
A-  $\frac{1}{4}$ " shoulder diameter,  $\frac{1}{2}$ " shoulder length, 10-24 thread shoulder bolt  
B- Oil embedded-thrust bronze bearing  
C- Needle Thrust Bearing, Bore  $\frac{1}{4}$ "



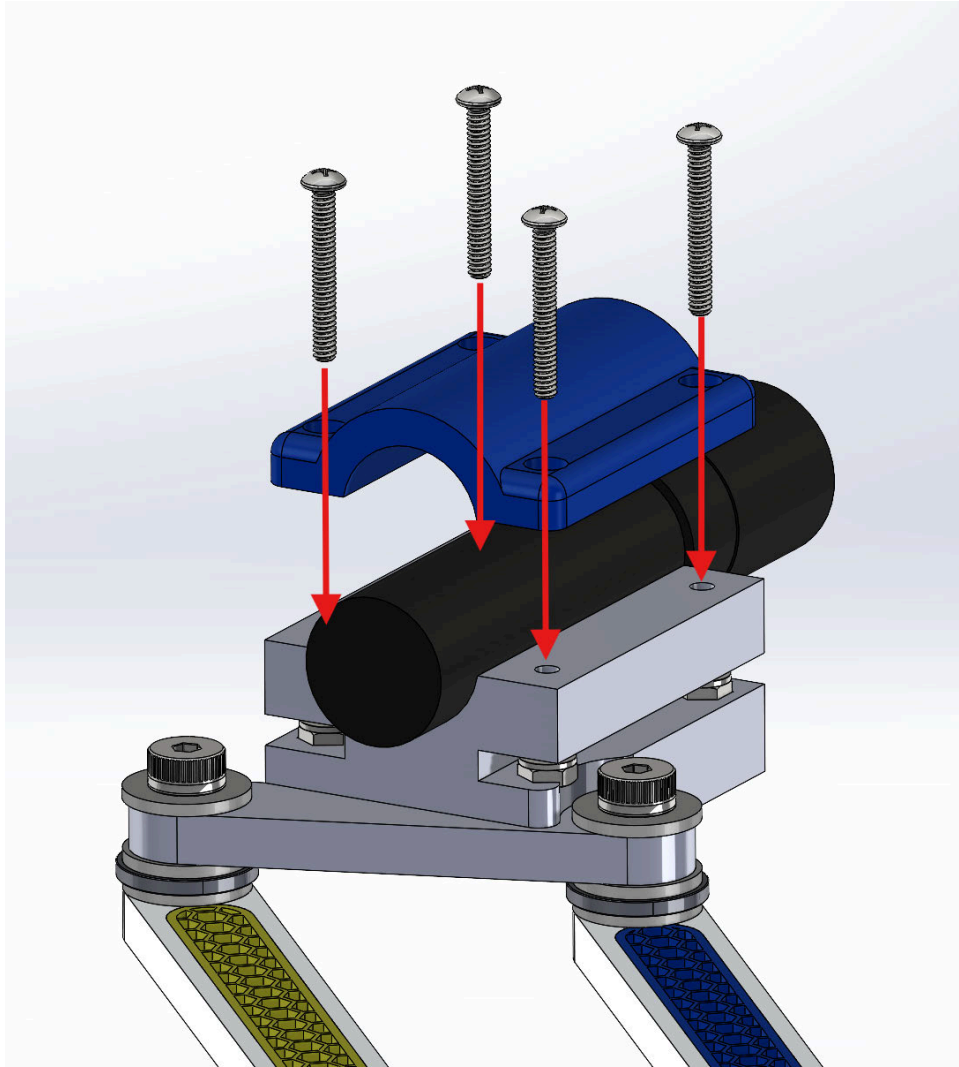


6. Create 4 stacks of washers and nuts and insert them into the slots in the coupler

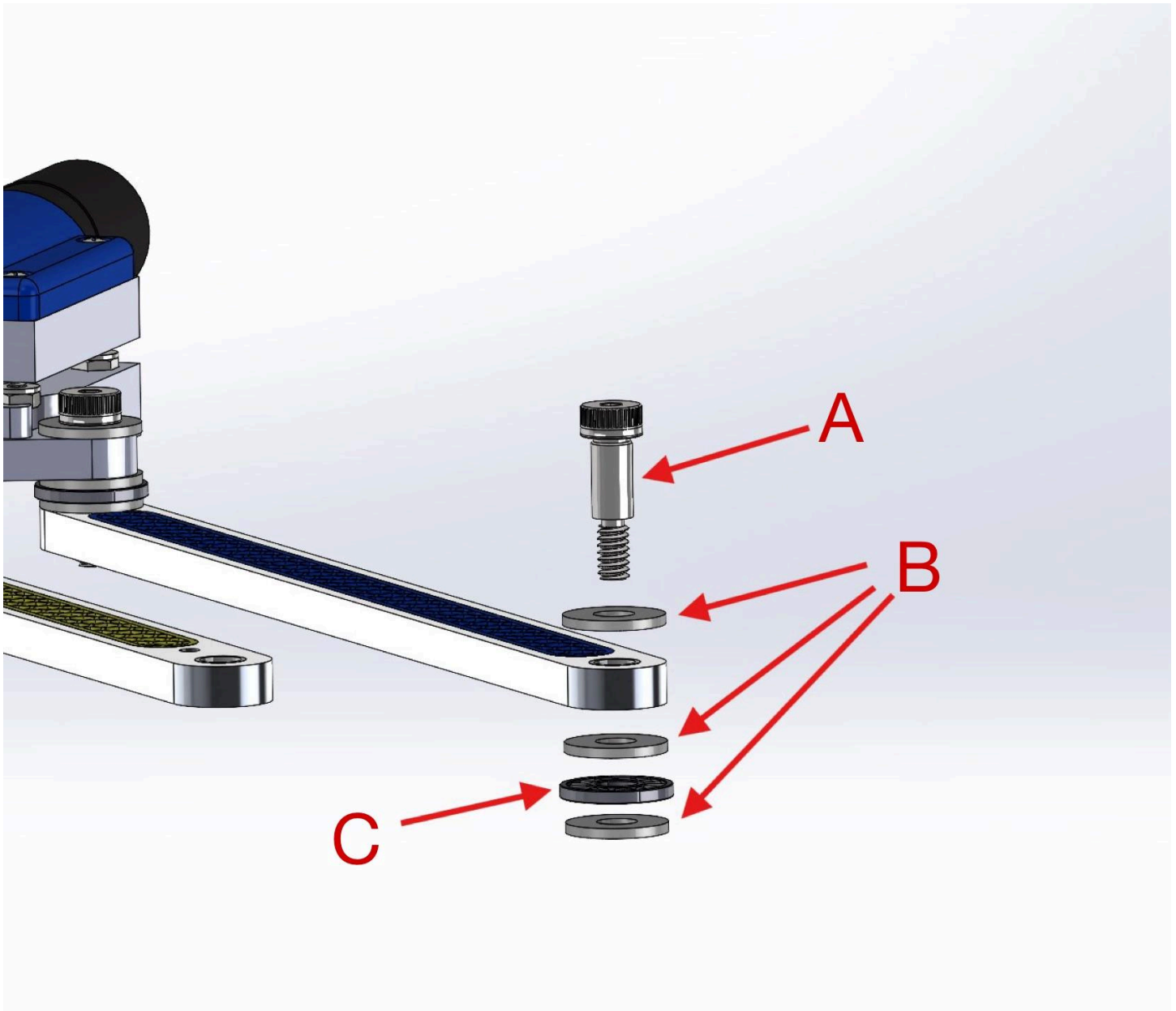




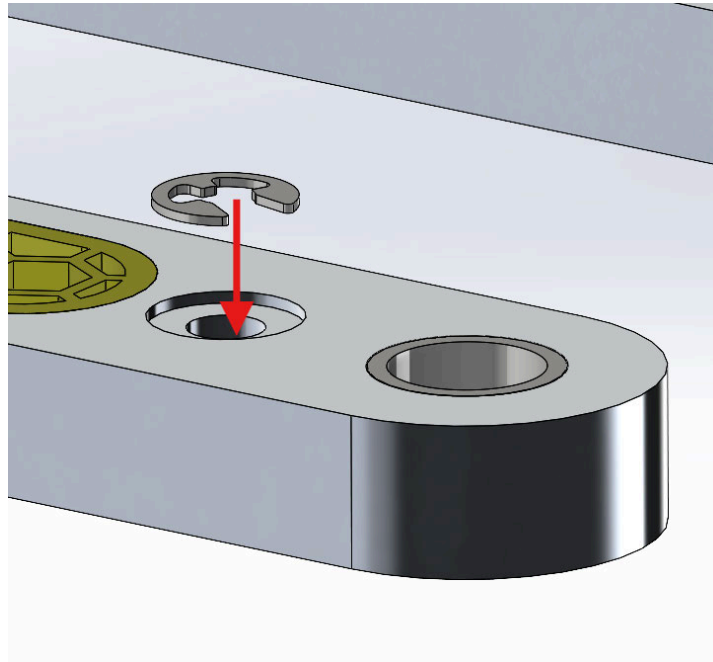
7. Screw Flashlight down onto the Flashlight Mount using stack of washer and nut using the passivated 18-8 stainless steel pan head phillips screw



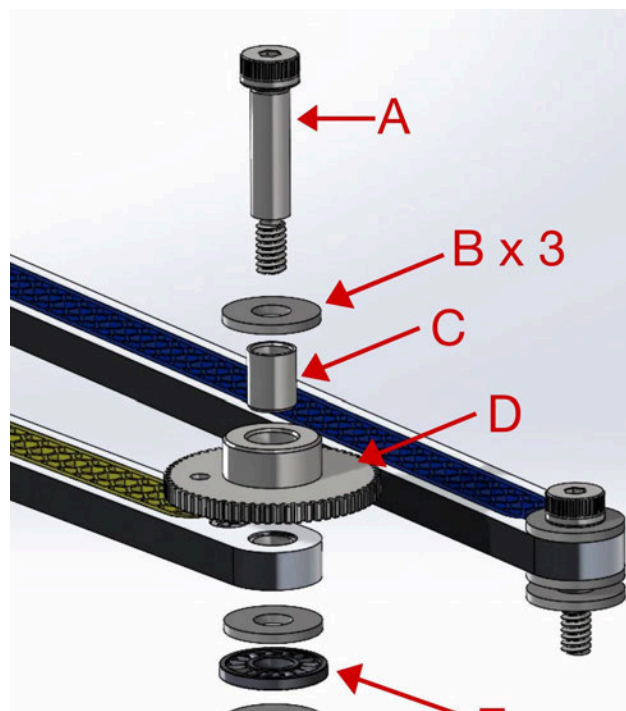
8. Create another joint assembly on the opposite end of the follower link
- A-  $\frac{1}{4}$ " shoulder diameter,  $\frac{1}{2}$ " shoulder length, 10-24 thread shoulder bolt
  - B- Oil embedded-thrust bronze bearing
  - C- Needle Thrust Bearing, Bore  $\frac{1}{4}$ "



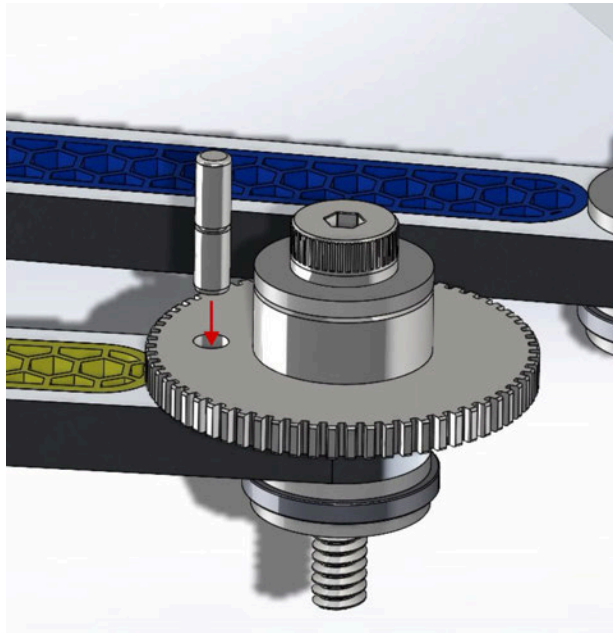
9. Press fit the E ring into the Input link



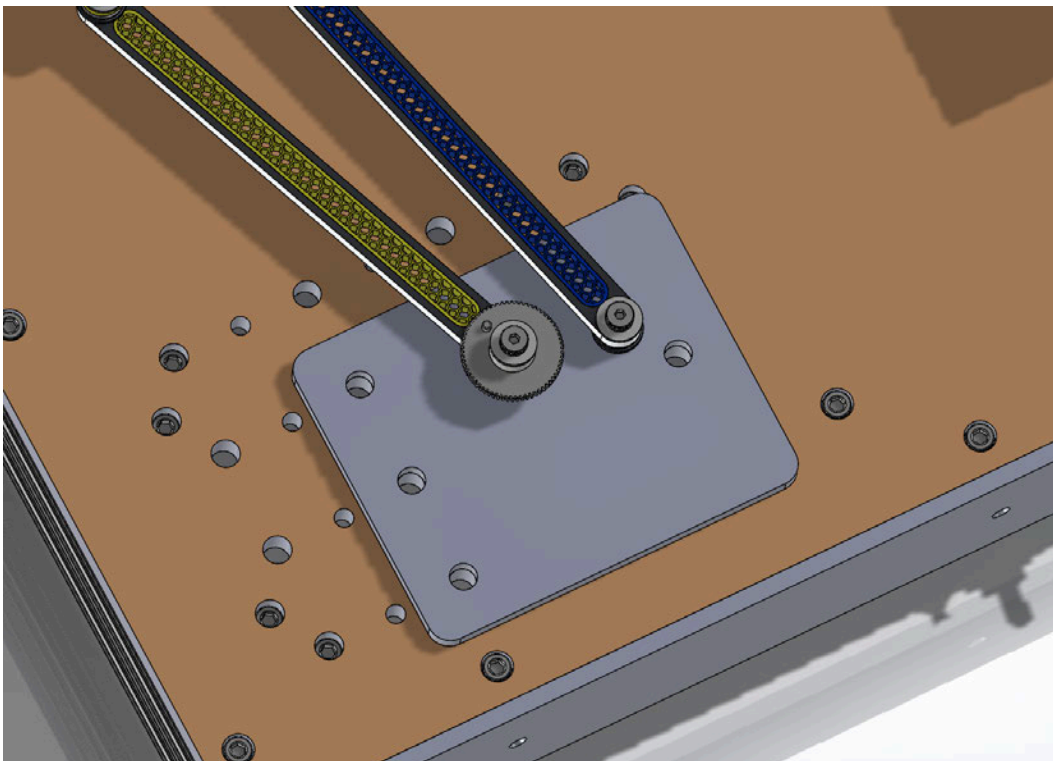
10. Create the new joint assembly on the input link using the gear
  - A-  $\frac{1}{4}$ " shoulder diameter,  $\frac{7}{8}$ " shoulder length, 10-24 thread shoulder bolt
  - B- Oil embedded-thrust bronze bearing
  - C- Plastic Bushing
  - D- 14.5 degree pressure angle plastic gear
  - E- Needle Thrust Bearing, Bore  $\frac{1}{4}$ "



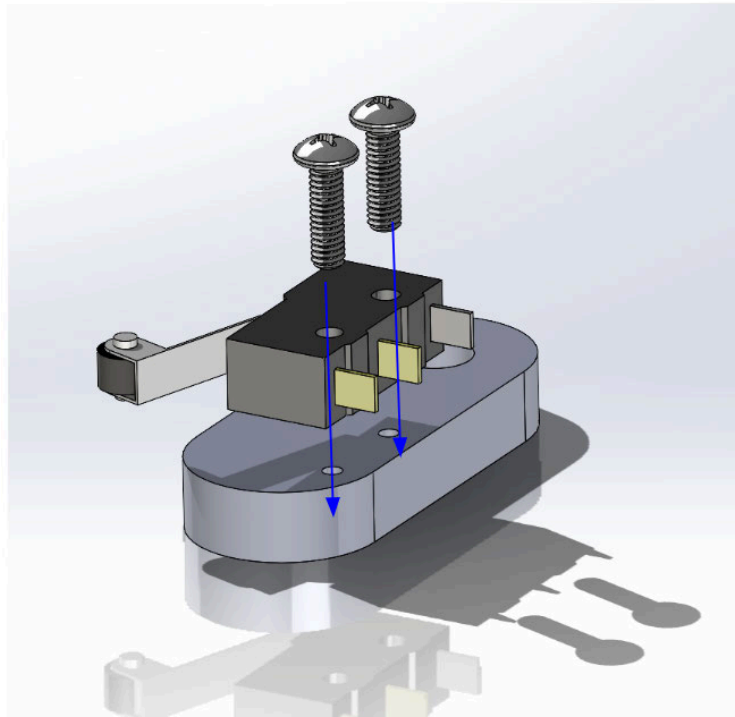
11. Press fit the dowel between the gear hole and the hole in the input link



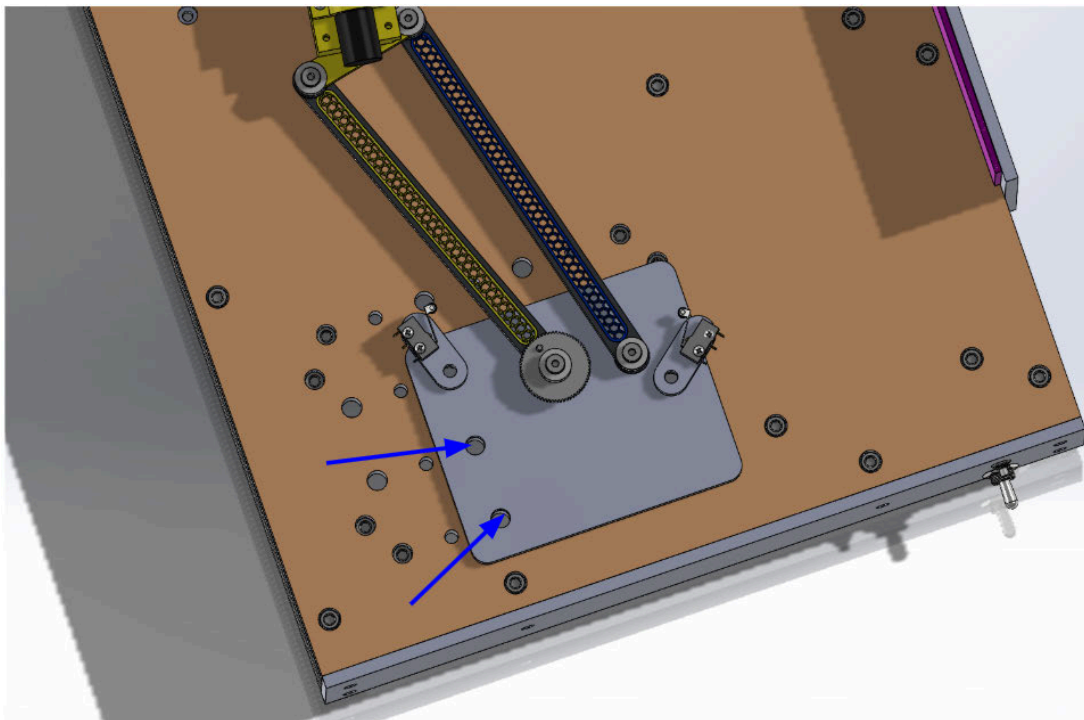
12. Attach to the baseplate



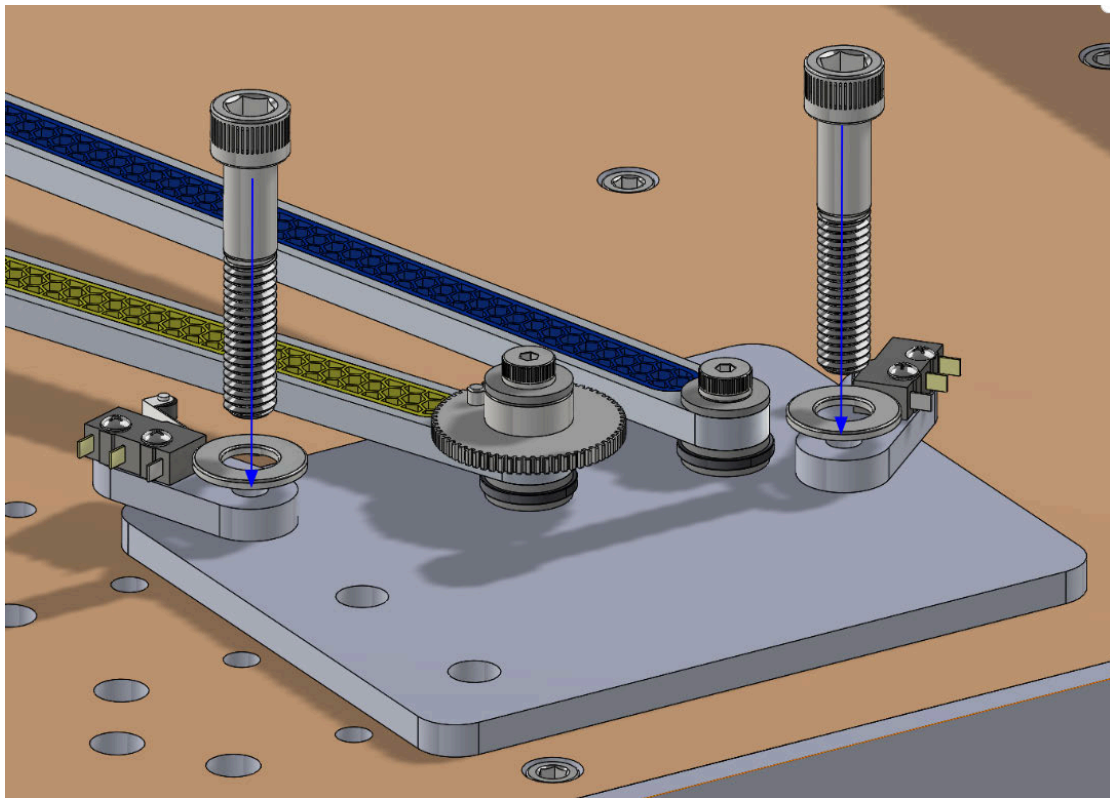
13. Create the hard stop assembly by attaching the limit switch to the hard stop using provided limit switch screws



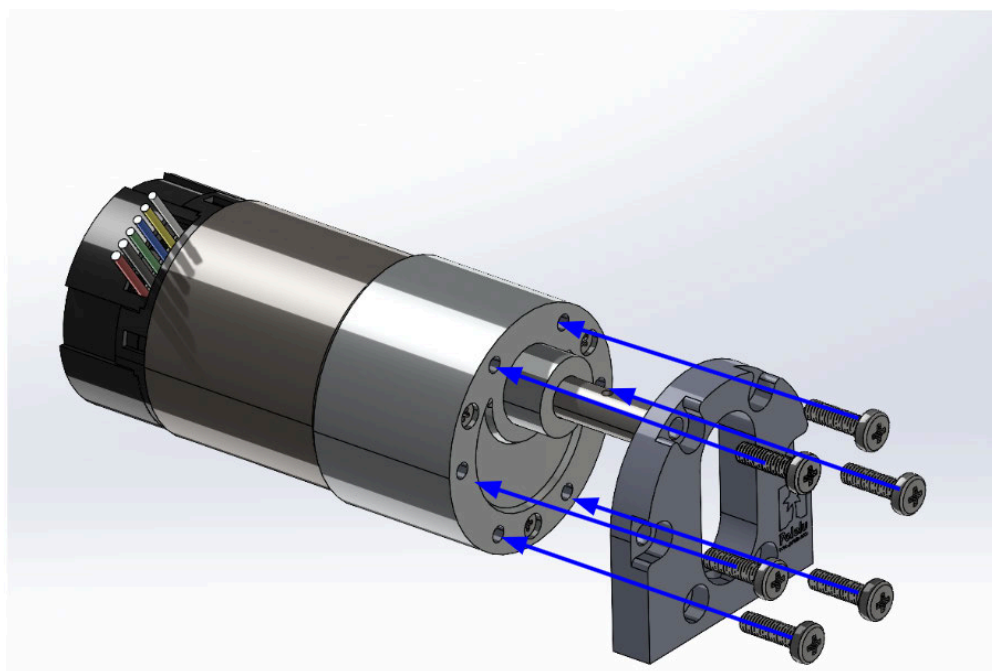
14. Set the baseplate on the playing field with Hard Stops aligned with the concentric holes of the base plate and playing field



15. Screw in the Hardstops with the shoulder bolts

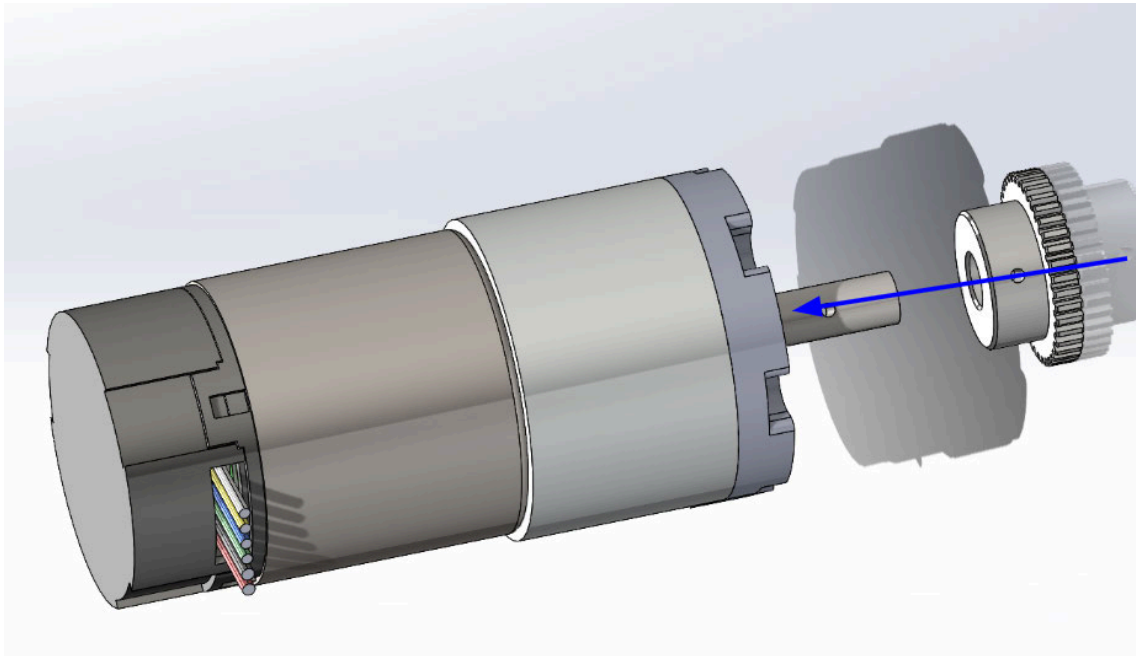


16. Attach the motor to the bracket

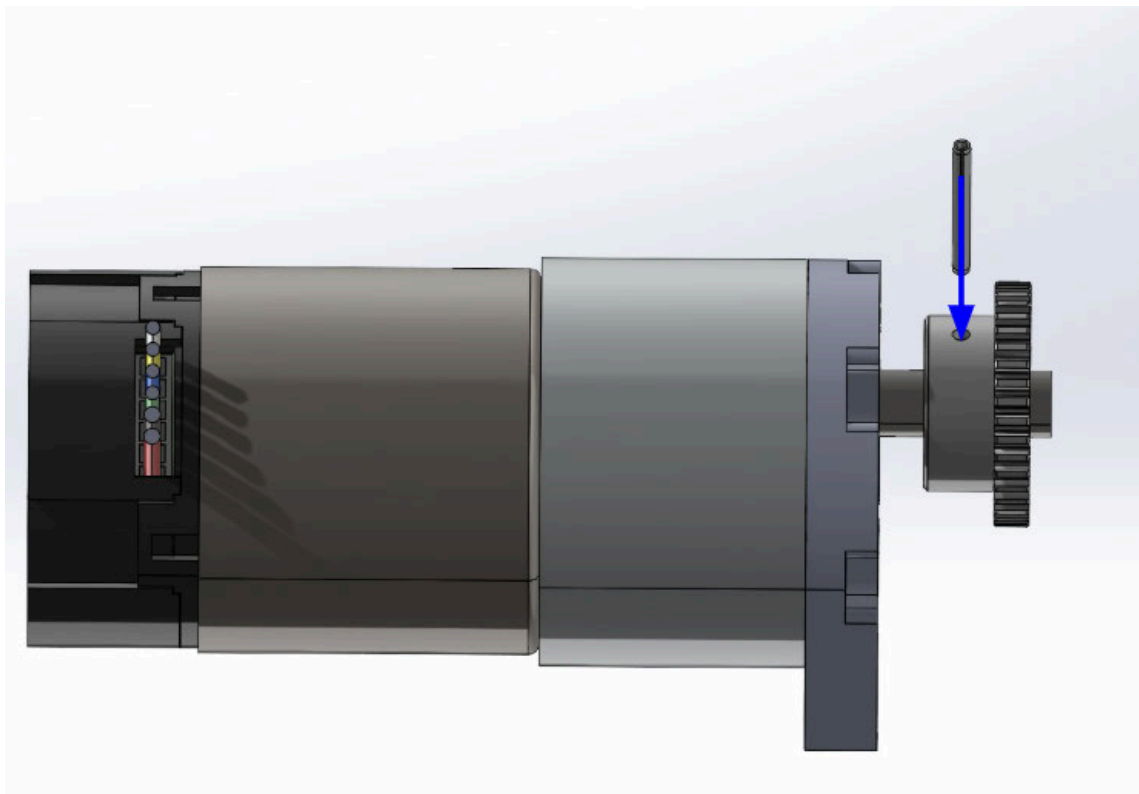




17. Press-fit the gear on

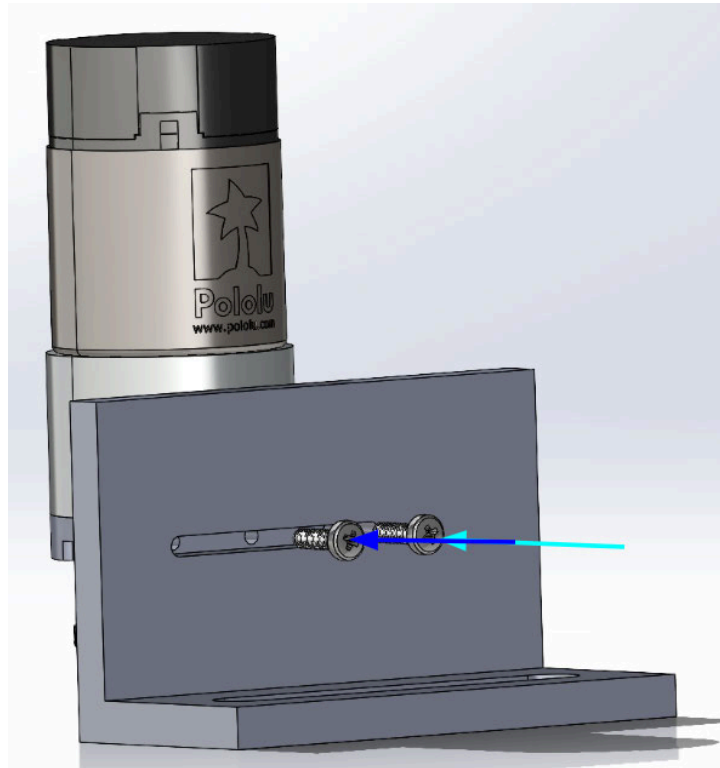


18. Insert the Spring pin in the concentric holes of the gear and motor output shaft

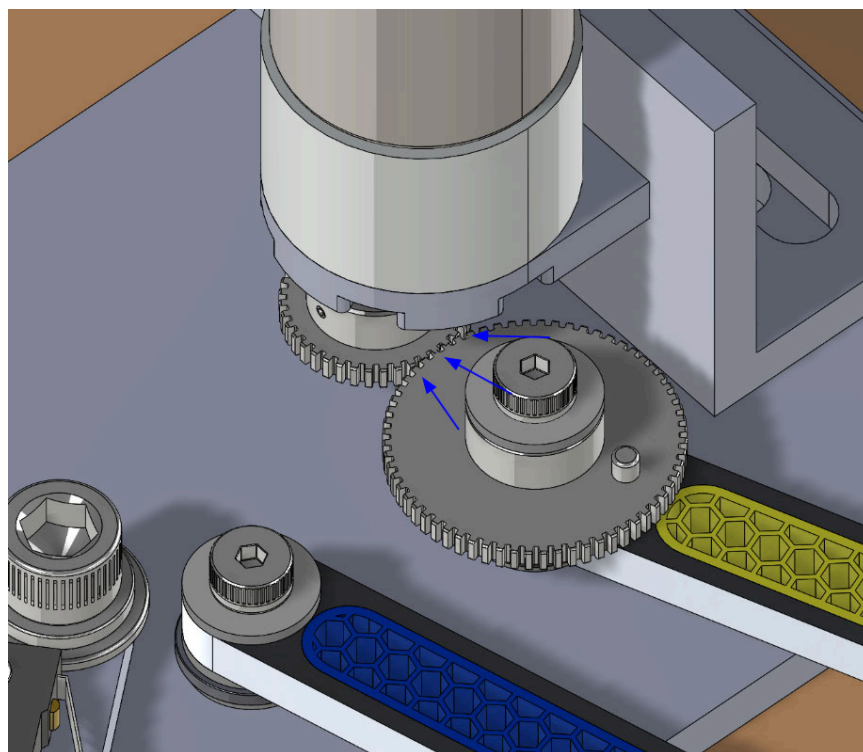




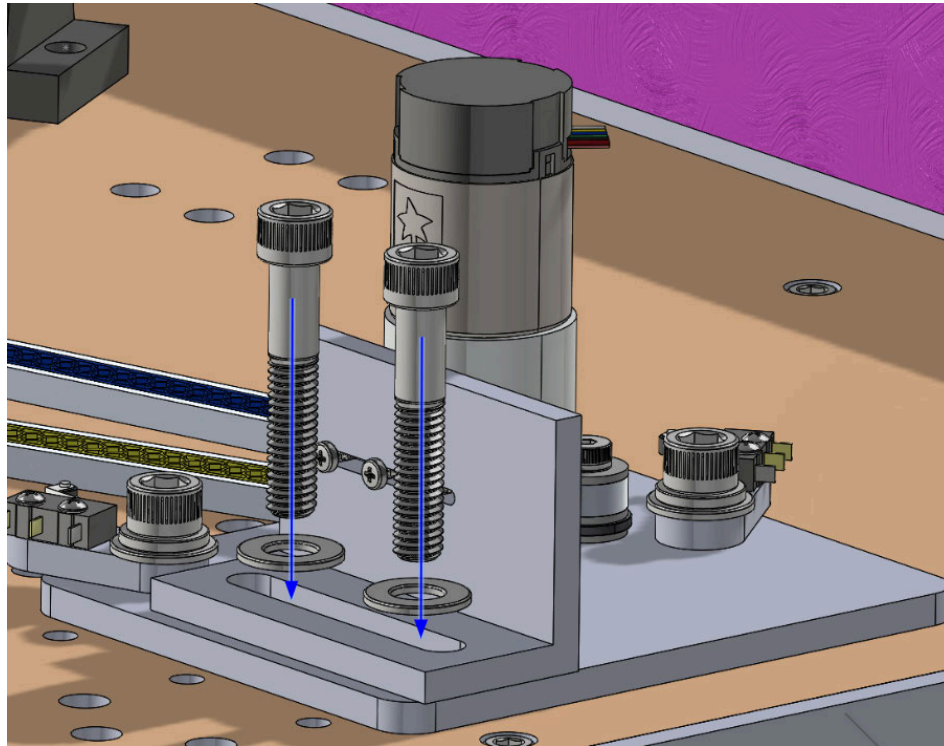
19. Screw the motor mount into the L-Bracket with washers



20. Align the motor assembly and attached output gear with the input link's gear



21. Fasten L-Bracket in the previously mentioned position, making sure to maintain constant meshing in the gears



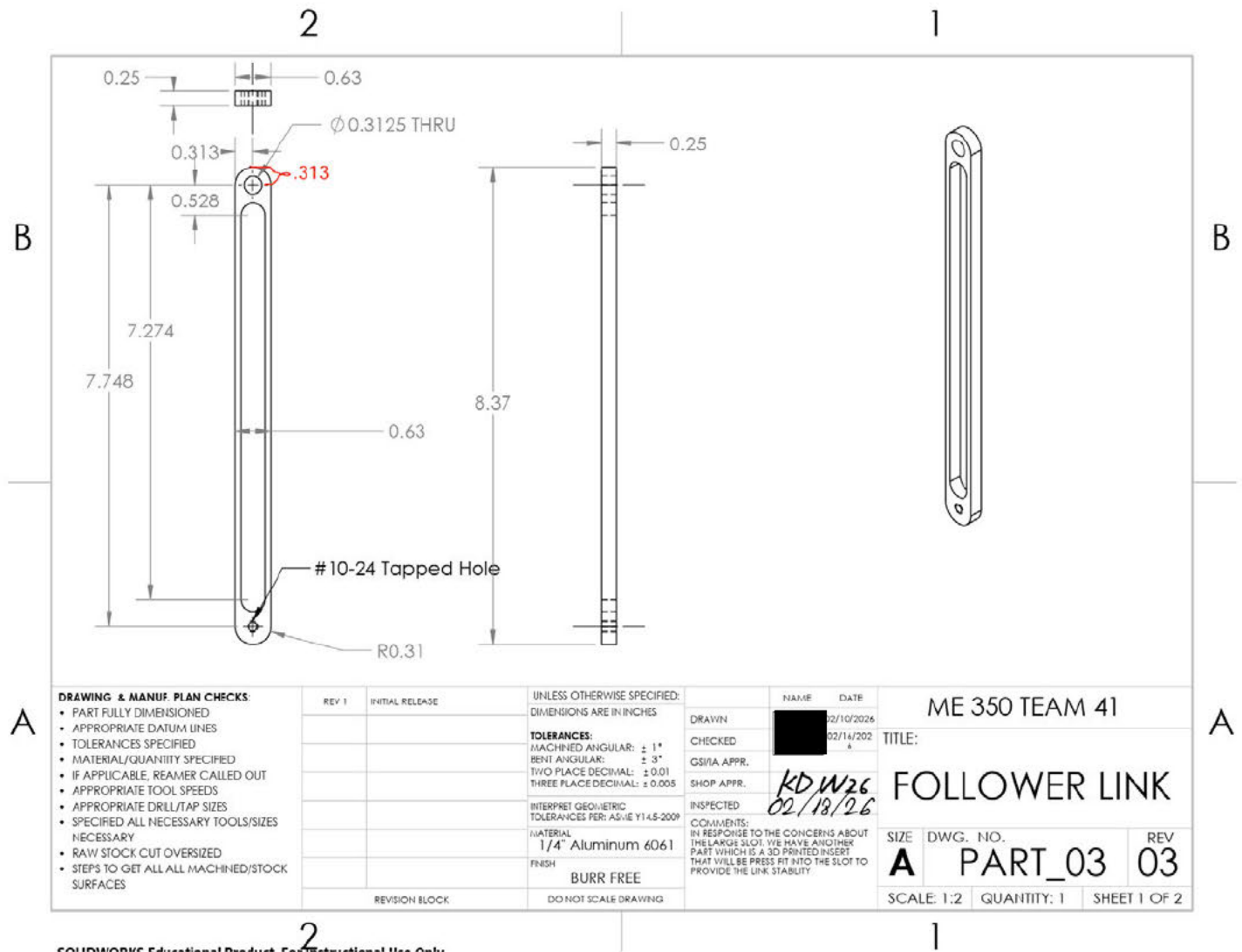
**Appendix B: Bill of Materials for the Final Design**

| Part No. | Part Title            | Material                | Dimension(s)                                    | Supplier      | Quantity |
|----------|-----------------------|-------------------------|-------------------------------------------------|---------------|----------|
| 1        | Input Link            | Aluminum                | 7.92" x 0.63" x 1/4"                            |               | 1        |
| 2        | Input Insert          | PLA                     | 6.37" x 0.42" x 1/4"                            |               | 1        |
| 3        | Follower Link         | Aluminum                | 8.37" x 0.63" x 1/4"                            |               | 1        |
| 4        | Follower Insert       | PLA                     | 7.17" x 0.42" x 0.25"                           |               | 1        |
| 5        | Coupler               | PLA                     | 2.44" x 2.76" x 0.97"                           |               | 1        |
| 6        | Flashlight Bracket    | PLA                     | 2.00" x 1.63" x 1/2"                            |               | 1        |
| 7        | Baseplate             | Aluminum                | 4.50" x 5.50" x 1/4"                            |               | 1        |
| 8        | L Bracket             | Aluminum                | 2" x 1.5" x 1/4"                                |               | 1        |
| 9        | Hard Stop             | Aluminum                | 1.68" x 0.68" x 1/4"                            |               | 2        |
| 10       | Dowel Pin             | Aluminum                | 1/8" Diameter x 1/2" long                       |               | 1        |
| 11       | Input Gear            | Nylon                   | 1.37" Diameter x 0.375" Thk                     | McMaster-Carr | 1        |
| 12       | Output Gear           | Nylon                   | 0.833" Diameter x 3/8"Thk                       | McMaster-Carr | 1        |
| 13       | 1/2" Shoulder screw   | Black-Oxide Alloy Steel | 1/4" Diameter x 1/2" Shoulder, 10-24 Thread     | McMaster-Carr | 3        |
| 14       | 7/8" Shoulder screw   | Black-Oxide Alloy Steel | 1/4" Diameter x 7/8" Shoulder, 10-24 Thread     | McMaster-Carr | 1        |
| 15       | Needle Thrust Bearing | Steel                   | Bore .250"                                      | McMaster-Carr | 4        |
| 16       | Bronze Thrust Bearing | Bronze                  | 1/4" Shaft Dia., 5/8" OD, 1/16" Thk             | McMaster-Carr | 12       |
| 17       | 1/2" Bushing          | Bronze                  | 1/4" Shaft Dia., 1/4" Lg                        | McMaster-Carr | 4        |
| 18       | 3/8" Bushing          | Bronze                  | 1/4" Shaft Dia., 3/8" Lg                        | McMaster-Carr | 1        |
| 19       | Limit Switch          |                         | 0.78" x 1/4" x 0.4"                             | Jameco        | 2        |
| 20       | #4-40 Pan Head Screws | Stainless Steel         | 7/8" Long x 4-40 Thread                         | McMaster-Carr | 4        |
| 21       | #4-40 Hex Nut         | Bronze                  | 1/4" long x 4-40 Thread                         | McMaster-Carr | 4        |
| 22       | Brass Washer          | Brass                   | 3/8"                                            | McMaster-Carr | 4        |
| 23       | E Ring                | 1060-1090 spring steel  | .095" Shaft Dia., .23" Outer Dia., .0125" Width |               | 1        |
| 24       | Pololu Motor Mount    |                         | 37D mm                                          | Pololu        | 1        |
| 25       | 30:1 Metal Gearmotor  |                         | 37Dx68L mm                                      | Pololu        | 1        |
| 26       | M3 Screws             |                         | 3mm diameter x 8mm long, M3                     | Pololu        | 9        |
| 27       | AR-100 LED Flashlight |                         | .874" Max Diameter x 3.7" Long                  |               | 1        |

|    |                          |                            |                                                 |               |   |
|----|--------------------------|----------------------------|-------------------------------------------------|---------------|---|
| 28 | Socket Head<br>Cap Screw | Black-Oxide<br>Alloy Steel | 3/8"-16 Thread Size, 2" Long                    | McMaster-Carr | 3 |
| 29 | 1/2" shoulder<br>screw   | Alloy Steel                | 1/4" Diameter x 1/2" Shoulder,<br>1/4-20 Thread |               | 2 |

| Purchased Part No. | Qty | Price/Unit | Supplier Part No. | Cost     |
|--------------------|-----|------------|-------------------|----------|
| 11                 | 1   | \$29.70    | 57655K27          | \$29.70  |
| 12                 | 1   | \$24.90    | 57655K21          | \$24.90  |
| 14                 | 1   | \$6.62     | 91259A541         | \$6.62   |
| 18                 | 1   | \$0.60     | 6391K33           | \$0.60   |
| 29                 | 2   | \$5.48     | 90807A113         | \$10.96  |
| Total Shipping     | 2   |            |                   | \$35.77  |
| Total              |     |            |                   | \$108.55 |

**Approval Packages:** For each part that underwent any kind of machining (including those purchased) for the linkage design (Gate Review 1), provide the (1) drawing, and (2) manufacturing plan.



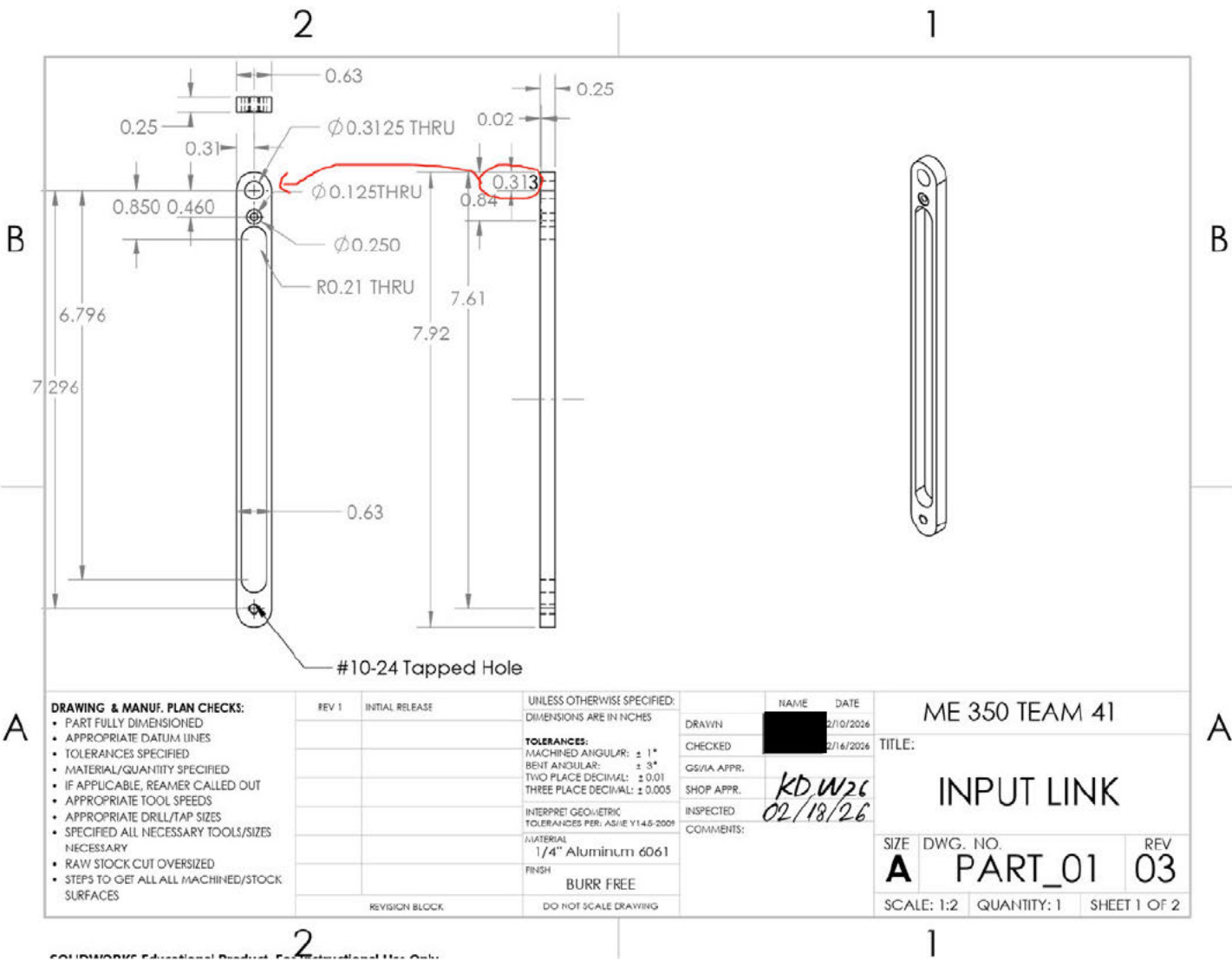
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MANUFACTURING PLAN

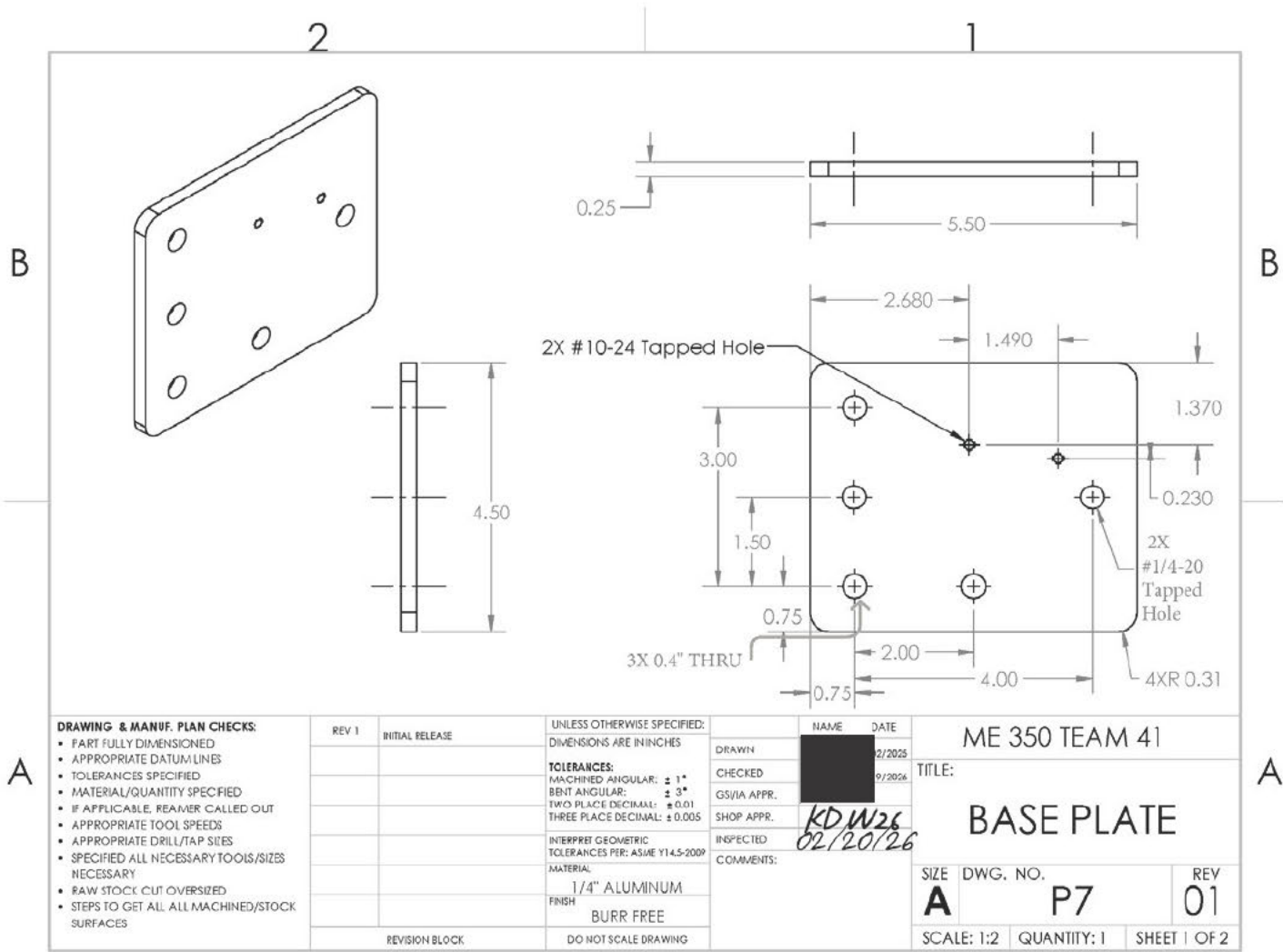
RAW MATERIAL STOCK: 1/4" aluminum 6061

| STEP | PROCESS DESCRIPTION                                 | MACHINE  | FIXTURE                                                | TOOL(S)                       | SPEED (RPM) |
|------|-----------------------------------------------------|----------|--------------------------------------------------------|-------------------------------|-------------|
| 1    | Cut out the main shape and slot with waterjet       | waterjet | N/A                                                    | N/A                           | N/A         |
| 2    | Find Datum                                          | Mill     | toe clamp fixture<br>Vice, Part Stop, 1.625" parallels | Edgefinder, Drill Chuck       | 1000        |
| 3    | Centerdrill the hole that will be tapped            | Mill     | Vice, Part Stop, 1.625" parallels                      | Drill Chuck, #2 centerdrill   | 1200        |
| 4    | Drill the initial hole to be tapped                 | Mill     | Vice, Part Stop, 1.625" parallels                      | Drill Chuck, #25 drill bit    | 1500        |
| 5    | Drill the THRU hole at the top                      | Mill     | Vice, Part Stop, 1.625" parallels                      | Drill Chuck, 19/64" drill bit | 1000        |
| 6    | Ream the hole at the top (bushing will be inserted) | Mill     | Vice, Part Stop, 1.625" parallels                      | 0.3115 reamer, 5/16" collet   | 100         |
| 7    | Remove and Deburr                                   | N/A      | N/A                                                    | Deburring Tools               | N/A         |
| 8    | Tap the tapped hole on EZ tap                       | N/A      | EZ-tap vice                                            | EZ-tap, 10-24 tap             | N/A         |
|      |                                                     |          |                                                        |                               |             |
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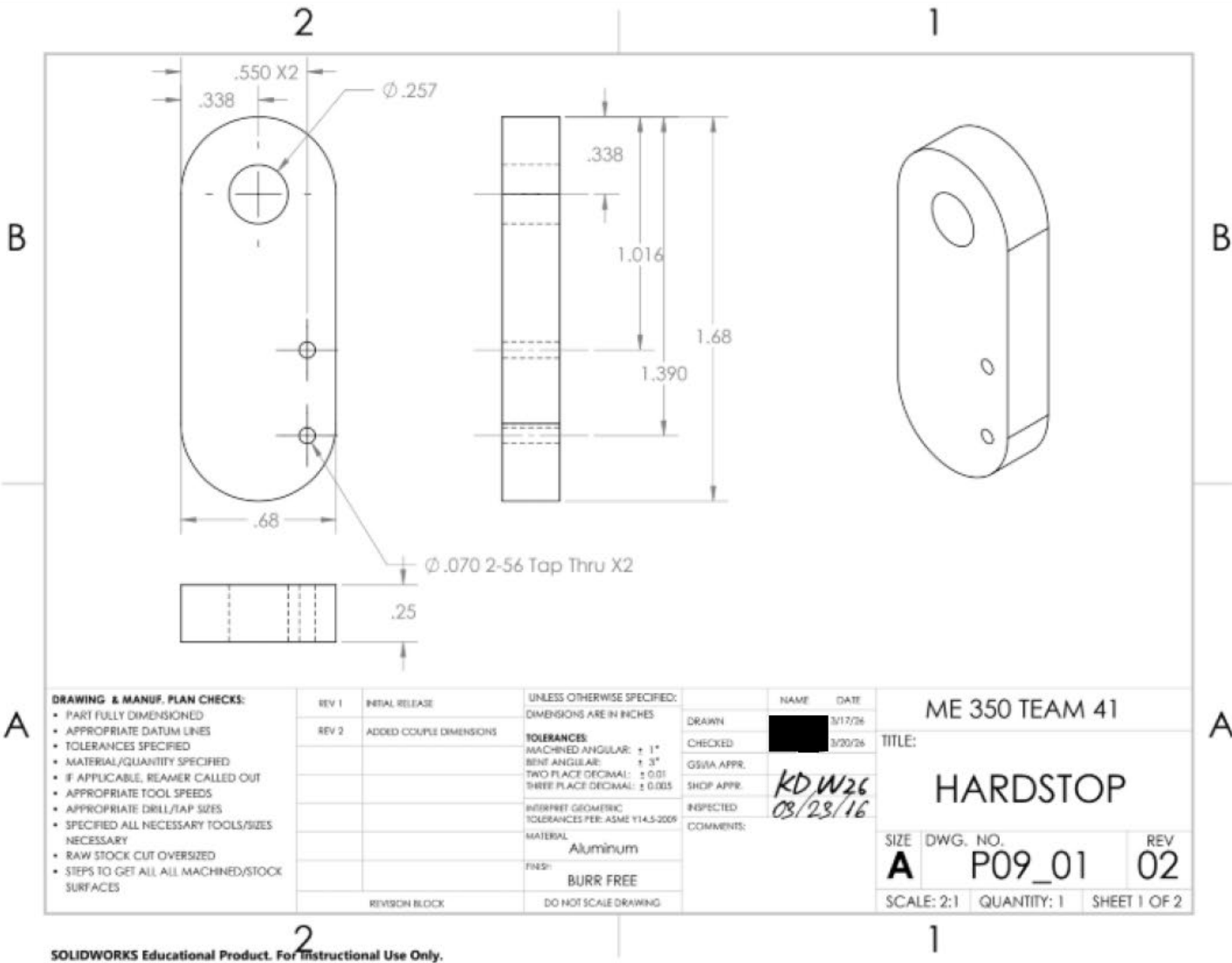
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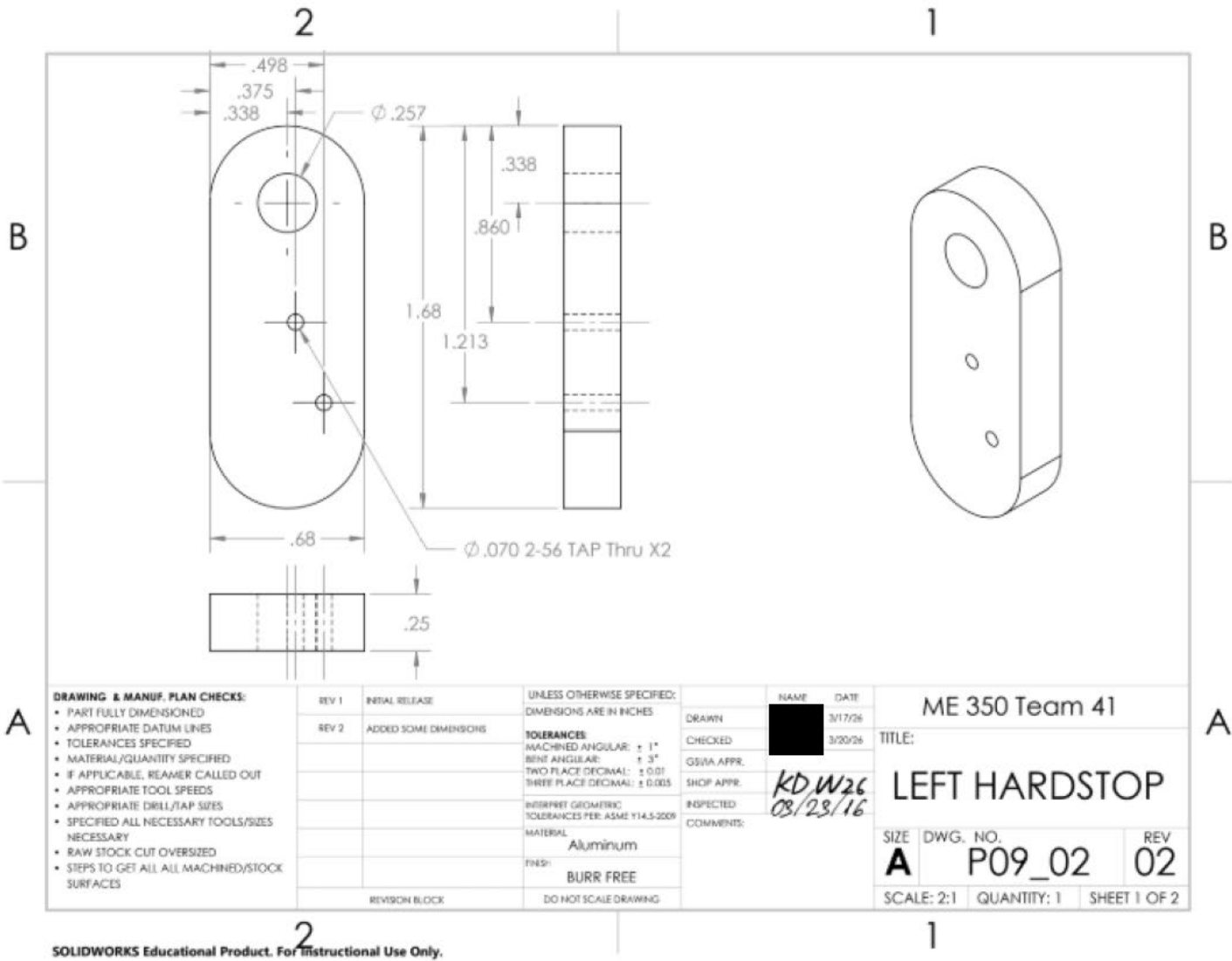
MANUFACTURING PLAN

RAW MATERIAL STOCK: 1/4" Aluminum Sheet

| STEP | PROCESS DESCRIPTION                            | MACHINE  | FIXTURE | TOOL(S)                                               | SPEED (RPM) |
|------|------------------------------------------------|----------|---------|-------------------------------------------------------|-------------|
| 1    | Cut outer profile                              | Waterjet |         |                                                       |             |
| 2    | Mount workpiece leaving at least 1/2' overhang | Mill     | Vise    |                                                       |             |
| 3    | Square workpiece                               | -        | Vise    | 5/16" endmill, 3/8" collet, 1 1/2" parallels          | 1200        |
| 4    | Set Datum (as in the drawing)                  | -        | Vise    | Edgefinder, 3/8" collet, 1 1/2" parallels             | 1000        |
| 5    | Center Drill the 0.40" Holes                   | -        | Vise    | #5 center drill, drill chuck, 1 1/2" parallels        | 1200        |
| 6    | Drill the 0.40" Holes                          | -        | -       | 13/32" drill bit, drill chuck, 1 1/2" parallels       | 600         |
| 7    | Center Drill the 1/4-20 Holes                  | -        | -       | 5 center drill, drill chuck, 1 1/2" parallels         | 1200        |
| 8    | Drill the 1/4-20 Holes                         | -        | -       | #7 drill bit, drill chuck, 1 1/2" parallels           | 1200        |
| 9    | Center Drill the 10-24 Holes                   | -        | -       | #3 center drill, drill chuck, 1 1/2" parallels        | 1200        |
| 10   | Drill the 10-24 Holes                          | -        | -       | #25 drill bit, drill chuck, 1 1/2" parallels          | 1500        |
| 11   | Tap the 1/4-20 Holes                           | -        | -       | 1/4-20 tap, tap wrench, drill chuck, 1 1/2" parallels |             |
| 12   | Tap the 10-24 Holes                            | -        | -       | 10-24 tap, tap wrench, drill chuck, 1 1/2" parallels  |             |
| 13   | Remove from Mill and Deburr                    |          |         | Deburring tools                                       |             |



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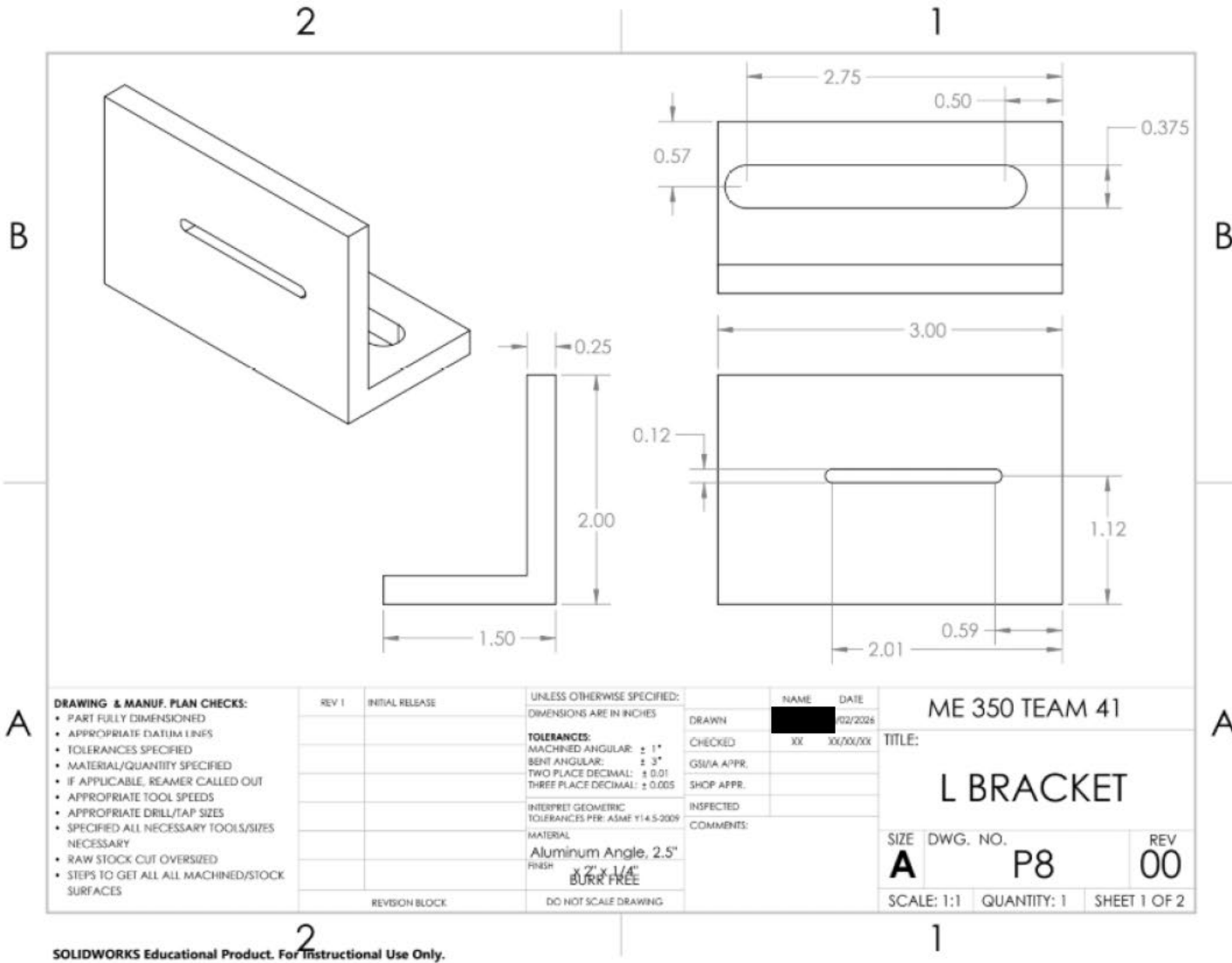
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SOLIDWORKS Educational Product. For Instructional Use Only.





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MANUFACTURING PLAN

RAW MATERIAL STOCK: XXXX

| STEP | PROCESS DESCRIPTION                                                   | MACHINE  | FIXTURE        | TOOL(S)                                                                   | SPEED (RPM) |
|------|-----------------------------------------------------------------------|----------|----------------|---------------------------------------------------------------------------|-------------|
| 1    | cut stock > 1/8" of final length then deburr                          | Band Saw | Vice           | Deburring tool                                                            | 300 ft/min  |
| 2    | Mount on mill with >1/8" overhang                                     | Mill     | Vice           | 1 1/2" Parallels                                                          |             |
| 3    | Face ends of parts to fully machined surface and debur                | Mill     | Vice           | 1 1/2" Parallels, 3/4" 2-flute endmill, 3,4 Collet                        | 500         |
| 4    | Install a workstop and zero, then remove the part, deburr and measure | Mill     | Vice, Workstop | Calipers, deburring tools                                                 |             |
| 5    | Bring part to final dimension 0.05" or smaller passes                 | Mill     | Vice, Workstop | 1 1/2" Parallels, 3/4" 2-flute endmill, 3,4 Collet                        | 500         |
| 6    | Find datums                                                           | Mill     | Vice, Workstop | 1 1/2" Parallels, 1/2" collet, edgefinder                                 | 1000        |
| 7    | Plunge Mill 5/16" Slot                                                | Mill     | Vice, Workstop | 1 1/2" Parallel, 1/4" end mill, 1/4" collet, 5/16" end mill, 5/16" collet | 500         |
| 8    | rotate part and plunge mill 1/8" slot                                 | Mill     | Vice, Workstop | 1 1/2" Parallel, 1/4" end mill, 1/8" collet, 1/8" end mill                | 500         |
| 9    | Remove the part and deburr                                            |          |                | Deburring Tools                                                           |             |
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